
How Effective is Green Regulatory Threat?

Empirical Evidence from Canadian Plant-Level Data

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Abstract

Governments are increasingly relying on environmental policies predicated on the assumption that the mere threat of regulation will entice companies to reduce their toxic emissions. This paper makes an attempt at identifying the effect of regulatory threat theoretically and empirically. The key to identifying regulatory threat's effect on the environment is to condition firms' response on their characteristics, namely, their *environmental exposure* that combines the effect of firm size, emission intensity, pollutant toxicity, and population at risk, and their *abatement ladder rung* that determines whether individual firms will participate in pollution abatement. Using 1993-99 data from the Canadian National Pollutant Release Inventory (NPRI), empirical analysis establishes that the theoretical predictions help identify statistically significant effects of regulatory threat that are, however, very small in magnitude. Green regulatory threat does not appear to be an effective instrument in Canada.

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1 Introduction

‘Green’ regulatory threat has become an important avenue for governments through which they pursue preemptive compliance with environmental standards without having to actually implement and monitor such standards. If threats lead to sufficient reductions in emissions, they may save considerable implementation and monitoring costs. However, green regulatory threat can only induce emission reductions when firms find the threat credible and do not free-ride on other firms’ abatement effort.

This paper is an attempt to model regulatory threat theoretically, and to identify its effect empirically by utilizing Canadian plant-level data on pollutant emissions between 1993 and 1999. Distinguishing between the effects from regulatory threat and the effects from other types of ‘voluntary’ abatement efforts is a key issue tackled in this paper. The tool through which this is accomplished is to condition firms’ environmental responses on their characteristics.

The nature of threats is usually such that they consist of a desired effort and a threatened sanction. For a threat to be effective, the expected present-discounted cost of the consequences must exceed the cost of the desired effort. In the context of green regulatory threat, a firm must make sufficient progress towards an environmental objective to avoid being sanctioned through regulatory intervention (e.g., an environmental standard or an emission tax). To induce the desired emission reductions, firms must be compensated through a lower probability of a regulatory intervention.

In addition to regulatory threat, there are other mechanisms that induce firms to reduce emissions without government intervention. One important such mechanism is green consumerism. Theoretical papers such as Arora and Gangopadhyay (1995) have hinted at the possibility that firms may overcomply with emission standards if they are subject to green consumerism and oligopolistic markets. In a recent paper (Antweiler and Harrison, 2003) we have explored the empirical relevance of green consumerism by conditioning a firm’s emission level on firm characteristics that can be linked to green consumerism. Based on predictions derived from a theoretical model, our empirical work found that environmentally-leveraged firms that are exposed to consumer markets have lower emissions, and are reducing emissions faster. For sectorally-diversified firms, the notion of “environmental leverage” captures a negative correlation between the intra-firm revenue share of a plant and its emission share. A firm is considered environmentally leveraged when a small unit with an above-average pollution intensity is “leveraging” a large unit with below-average pollution intensity. When consumers lack the ability to target individual units of a firm, firms that are environmentally leveraged will respond more intensely to green consumerism.

In this new paper I am interested in identifying the effect of regulatory threat, and again it is necessary to condition a firm’s abatement effort on

characteristics that can be linked to regulatory threat. Governments care about the *environmental exposure* of firms, three firm characteristics determine their abatement effort: (1) the volume of emissions (which combines output size with pollution intensity); (2) the toxicity of emissions; and (3) the extent to which these emissions put population at risk (which is determined by the firm's location and the size of the population in the surrounding area). These firm characteristics alone are not sufficient to determine a firm's response to regulatory threat. Firms' abatement effort is also conditioned on whether they meet participation and incentive constraints determined by their unabated pollution intensity and unit abatement cost. Then firms' position relative to other firms—their rung on the *abatement ladder*—determines their participation in abatement activity.

The model will show that a combination of a firm's *environmental exposure* and *abatement ladder rung* provide a suitable basis for backing out the effect of regulatory threat empirically. This new paper also provides a rigorous theoretical basis for the *prima facie* observations discussed in Harrison and Antweiler (2001). There we found only scant evidence for the effect of regulatory threat by looking at differential responses between firms subject to different types of scheduled or threatened regulation under the Canadian Environmental Protection Act (CEPA).

Green regulatory threat is one form of induced ('voluntary') industry self-regulation. It is by no means the only such form. The recent literature has increasingly focused on public voluntary agreements (PVA) in which firms commit to specific (environmental) targets and are supported by governments through a subsidy, but where membership in a PVA is not mandatory.¹ Wu and Babcock (1999) consider the efficiency of PVAs relative to explicit regulation, stating conditions under which PVAs are more advantageous. However, when Lyon and Maxwell (2002) compare the efficiency of PVAs and regulatory threat, they find that PVAs are not always welfare-enhancing. Segerson and Miceli (1998) compare the effectiveness of regulatory threat (the "stick approach") relative to cost-sharing subsidies (the "carrot approach") for inducing participation in PVAs. Their results indicate that a multitude of factors play a role (notably the allocation of bargaining power), and that the environmental outcome is ambiguous.

Green regulatory threat may also contribute to lower emissions by inducing firms to invest in innovation. Cadot and Sinclair-Desgagné (1995) provide a theoretical analysis where firms and governments bargain over the time for adopting stricter environmental standards. Firms argue that they require time to innovate environmental technology in order to comply with future standards. Imposing a standard at once may be suboptimal, and regulatory threat may provide an adequate incentive scheme to ensure firms' environmental innovation.

¹The subsidy is often not pecuniary. PVAs also differ in intensity; some take the form of legally binding contracts, while others are mere statements of intention.

Methodologically, the work in this paper is related to papers such as Arora and Cason (1999), who combine data from a toxic release inventory with demographic information through geographic matching. Where Arora and Cason (1999) are concerned with the location choice of manufacturing facilities and the linkage to community characteristics, here the linkage to spatial data is used to back out the potential effect of regulation and regulatory threat.

The remainder of the paper is structured as follows. In section 2 I develop a rudimentary theory of regulatory threat, with a particular focus on the role of participation and incentive constraints that determine which plants do or do not engage in abatement when regulation is introduced. Section 3 previews the data used in the empirical analysis and discusses the regulatory context in Canada. Section 4 focuses on the environmental exposure measure that plays a key part in the empirical analysis. This section also provides exposure rankings for pollutants and plants. Section 5 introduces the empirical implementation and discusses econometric issues relating to the estimating equations. Results are provided in section 6, and section 7 concludes.

2 Theory

The theoretical model strives to identify the firm characteristics through which the effect of regulatory threat can be identified. There are important limitations to this modeling approach. It ignores interactions between firms related to the free-rider problem. The principal economic agents in the model are firms and the government. Consumers only play a role as voters and ‘consumers’ of pollution. To focus on the essential features of the government-firm interactions, price and income effects are excluded.²

Consider a simple political economy model with N consumer-voters (indexed i) and J single-plant firms (indexed j). Consumers are exposed to K pollutants (indexed c). The physical distance between consumer i and firm j is given by d_{ij} .

The decision making proceeds in two stages. First, the government determines its optimal regulatory intervention for the case that regulation is indeed implemented. Simultaneously, it announces how likely it will implement such regulation depending on an environmental exposure measure that firms can influence by voluntary abating emissions. Second, firms determine their optimal response to the threatened intervention.

²This may not be as unrealistic as it sounds. Abatement expenditures are usually a relatively small portion of overall costs, and regulators may not take price and income effects into account.

2.1 Geography, Emissions, and Exposure

Plant emissions are linked to a household's environmental exposure through an attenuation function that is governed by the distance between plant and household. Unlike greenhouse gases where exposure is essentially global and the location of an emission source is unimportant, the reverse is true for toxic pollutants where exposure is local or regional.

Emissions are linked to exposure through two channels: *attenuation*, which translates emissions into ambient emission concentrations, and *toxicity*, which translates emission concentrations into health effects (actual or risk). For toxicity we assume that policy makers attach a weight κ_c to a pollutant, and we approximate attenuation by geographical proximity.

Consumer i 's *environmental exposure* X_i is the weighted sum of emissions $Z_j \equiv \sum_c \kappa_c Z_{jc}$ from plant j , attenuated by a function $E(d_{ij}; Z_j)$ that depends on the distance d_{ij} between consumer and plant. Hence,

$$X_i \equiv \sum_j E\left(d_{ij}; \sum_c \kappa_c Z_{jc}\right) \quad (1)$$

To limit the complexity of the model, I assume that attenuation occurs in a particularly simple form: consumers within radius r of a firm are considered exposed, and those beyond r are considered unaffected. Let $\mathcal{D}_i(r) \equiv \{j | d_{ij} \leq r\}$ and $\mathcal{D}_j(r) \equiv \{i | d_{ij} \leq r\}$ define, respectively, a set of all firms in consumer i 's vicinity, and a set of all consumers in firm's j 's vicinity. Thus,

$$X_i = \sum_{j \in \mathcal{D}_i(r)} \sum_c \kappa_c Z_{jc} \quad (2)$$

Further let $n_j(r) \equiv |\mathcal{D}_j(r)|$ denote the number of exposed consumers to emissions from plant j .³ The total economy-wide exposure is thus:⁴

$$X \equiv \sum_i X_i = \sum_i \sum_{j \in \mathcal{D}_i(r)} \sum_c \kappa_c Z_{jc} = \sum_c \kappa_c \sum_j n_j(r) Z_{jc} \quad (3)$$

³This implies a population density of $d_j(r) \equiv n_j(r)/(\pi r^2)$ around plant j .

⁴This is admittedly a simplistic exposure measure. First, it is a contemporaneous measure that neglects the cumulative effect of emissions. Second, it ignores the possibility that exposure is intrinsically non-linear. Limited exposure may have no health impact, while higher exposure rates may increase health risk more than proportional to the exposure concentration. Third, an economy-wide exposure measure implies that policy makers are willing to make inter-regional trade-offs: a higher exposure in one region may be compensated by a lower exposure in another region. The first concern can be addressed by cumulating exposure over time, and discounting past exposure with an appropriate decay rate. The last two concerns can be addressed by introducing appropriate non-linear modifications of equation (3). To preserve analytic tractability these options are not explored further. Addressing the cumulative effect will become important once sufficient longitudinal data are available; currently the NPRI covers less than a decade.

2.2 The Government

The government's role is to threaten K menus $(\bar{z}_c, \tau_c)$ of intervention, consisting of emission standard $\bar{z}_c$ and penalty rate τ_c for pollutant c .⁵ When firms exceed $\bar{z}_c$, they are required to pay a penalty (ie, an emission tax) of τ_c per unit of emission. Two familiar special cases are nested within menu $(\bar{z}_c, \tau_c)$: a simple emission tax $(0, \tau_c)$, and conventional command-and-control regulation $(\bar{z}_c, \infty)$. Considering a menu is a reflection of actual regulatory practice: policy makers often pursue a measure of *fairness*⁶ in addition to economic *efficiency*. While efficiency arguments unequivocally favour emission taxes over emission standards, firms—in particular those with low emission intensities—will appeal to governments' sense of "fairness" to implement standards that will let them escape penalties or taxes.⁷

Typically, governments will not regulate all pollutants but only those that impact large numbers of citizens. In determining regulatory intervention, the government must balance two types of costs: health costs that are a function of emissions X , and the abatement costs A borne by firms. The latter also includes the penalties for exceeding emission standards.⁸ When it has decided to pursue regulatory intervention, the government's objective is to choose the K menus of $(\bar{z}_c, \tau_c)$ in order to minimize the sum G^R of these costs:

$$G^R = \gamma \sum_i X_i(z, \tau) + \sum_j A_j(z, \tau) + F \quad (4)$$

Here γ is a weight that the government attaches to the health costs, and F is the government's cost of monitoring and enforcing the regulatory regime. 'Green' governments will choose a high γ . We can rewrite (4) using equation (3):

$$G^R = F + \sum_c \sum_j [\gamma \kappa_c n_j(r) Z_{jc}(z_c, \tau_c) + A_{jc}(z_c, \tau_c)] \quad (5)$$

Governments can pursue regulatory threat as an alternative to regulation.

⁵In the Canadian regulatory practice, emission taxes have not been used directly. Instead, courts have played a major role in assessing punitive damages when plants have violated environmental regulations. Courts take into account the severity of the breach, and thus implicitly impose an 'emission tax' that varies with the size of the excess emission. For modeling purposes an emission tax may thus not be entirely unrealistic. Stranlund (1995) explores regulatory regimes where noncompliance is sanctioned by fines.

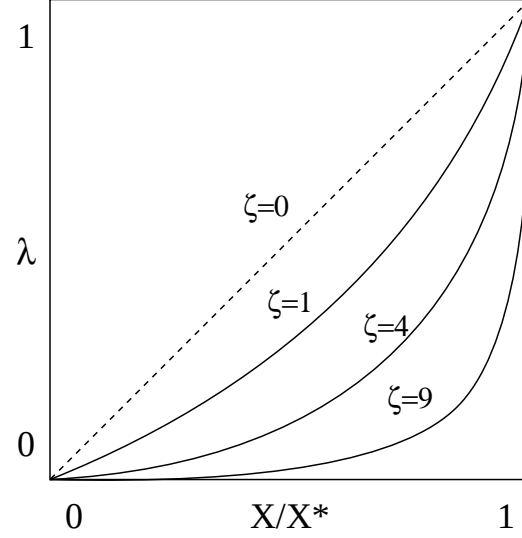
⁶Of course, there is no precise economic definition of "fairness," and thus different policy makers will give this term very different meaning. For a discussion of the relative merits of emission standards versus emission taxes see Harford and Ogura (1983).

⁷In the context of toxins, emission trading has not been proposed as an alternative. Unlike greenhouse gases, the market for toxin-specific permits would likely be small and suffer from liquidity problems and other frictions.

⁸Governments tend to care about the political cost of implementing an intervention menu. While a simple emission tax $(0, \tau_c)$ is more efficient than a simple emission standard $(\bar{z}_c, \infty)$, a combination of emission standard and emission tax will be politically less "costly" than a simple emission tax because the sum $\bar{z}_c \tau_c$ is not transferred to the government.

Figure 1: The Government's Incentive Scheme

The government announces an incentive scheme to firms through which they can influence the government's probability λ of enacting regulation. This function is determined by parameters X^* (the level of environmental exposure that triggers regulation with certainty) and ζ (the reluctance-to-regulate factor). The higher the reluctance factor ζ , the more "non-linear" the government's response. When $\zeta \rightarrow \infty$, the incentive scheme becomes a pure threshold function.



In particular, they can threaten to implement the regulatory regime described above with probability $0 < \lambda < 1$. If a government pursues a regulatory threat regime, it will minimize

$$G^N = \sum_c \sum_j [\gamma \kappa_c n_j(r) Z_{jc}(\lambda) + A_{jc}(\lambda)] \quad (6)$$

Regulatory threat is preferred to regulation when $\exists \lambda : G^N(\lambda) < G^R$. Since threatening regulation will always be less effective environmentally than actual regulation, a necessary condition for regulatory threat to be a viable option is that the monitoring and enforcement cost F of regulation is relatively large. Put another way, through regulatory threat governments can trade off administration costs for environmental welfare.

For regulatory threat to have an effect on firms' decision about pollution abatement, firms must be able to influence the government's probability λ of implementing regulation. For this purpose the government must announce an incentive scheme $\lambda = \Lambda(X)$ that links the implementation probability λ to the level of environmental exposure X . The following choice of a functional form for Λ is particularly flexible:⁹

$$\lambda = (X/X^*)^{1+\zeta} < 1 \quad (7)$$

Equation (7) expresses a number of desirable properties; it is illustrated in figure 1. Just like γ , X^* and ζ are political choices exogenous to this model.¹⁰

⁹Equation (7) is an ad-hoc choice that captures desirable empirical properties; it is not derived from first principles. This ad-hoc choice leaves two questions unanswered. First, what is the process through which the incentive scheme is determined? Second, what is the "optimal" incentive scheme?

¹⁰There is a growing literature that models the role of competing lobbies in determining such political choices. Maxwell et al. (2000) is a prime example of modeling green lobbies.

First, X^* denotes an upper threshold for the level of environmental exposure that will trigger regulation with certainty. This can be viewed as a government's re-election constraint. Second, $\zeta > 0$ measures the political reluctance to implement regulation. As ζ increases, the level of non-linearity increases and Λ turns increasingly into a threshold function. Third, the convexity of Λ ensures that firms face diminishing returns to voluntary pollution abatement (ensuring the existence of an interior solution). Governments with a low X^* and/or low ζ can be considered as environmentally friendly. On which segment of the Λ schedule will governments and firms operate: the flat or the steep part? As will become apparent below, the interesting part of the schedule is the flat part where λ is relatively small.

2.3 The Firm

Consider a firm j that generates output q_j . Generating one unit of q_j generates z_{jc} units of unabated emission of the $c = 1, \dots, K$ known pollutants. The firm is subject to regulatory threat consisting of a menu $(\bar{z}_c, \tau_c)$ of emission standard $\bar{z}_c$ and penalty τ_c if the standard is exceeded (ie, there are excess emissions).

Let π^R denote the firm's current-period profit under regulation, and let π^N denote the firm's profit without regulation. Firm j can invest θ_{jc} into abating emissions of pollutant c (per unit of emission), which reduces emissions to $\exp(-\theta_{jc}/b_{jc})$ times its unabated level z_{jc} . Here, b_{jc} is a cost factor per unit of emission. The firm's contemporaneous profit without regulation is

$$\pi_j^N = q_j \left[p - c_j - \sum_c z_{jc} \theta_{jc} \right] \quad (8)$$

Without threat of regulation, the obvious choice of θ_{jc} is zero. However, when regulation is threatened firms can influence the implementation probability by voluntarily abating emissions. The profit maximization problem will then become an intertemporal value maximization problem that we will explore in section 2.6.

Under regulation, the contemporaneous profit stream depends on the intervention menu $(\bar{z}_c, \tau_c)$:

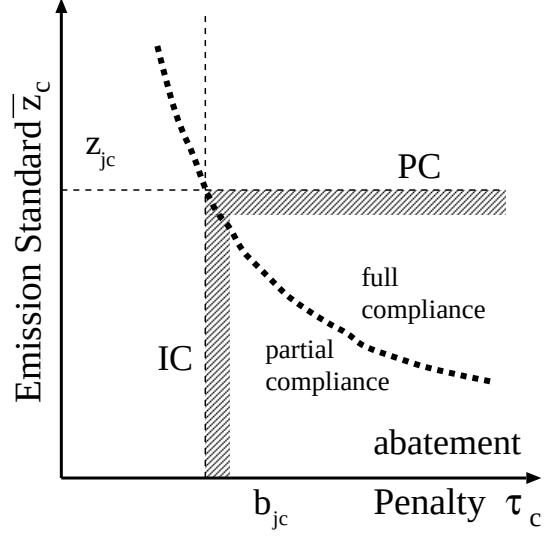
$$\pi_j^R = q_j \left[p - c_j - \sum_c \left[z_{jc} \theta_{jc} + \tau_c \max \left\{ z_{jc} e^{-\theta_{jc}/b_{jc}} - \bar{z}_c, 0 \right\} \right] \right] \quad (9)$$

It is important to acknowledge the simplifying assumptions present in this model. First, firms react to regulatory threat through abatement effort, but they do not lower their scale of production. Second, abatement effort does not reduce profitability so as to induce exit. Third, short-term relocation costs are prohibitive and thus firms do not change location.

The firm faces a participation constraint (PC) $z_{jc} > \bar{z}_c$: abatement activity will only commence when the firm's unabated pollution intensity z_{jc}

Figure 2: The Firm's Participation Constraint, Incentive Constraint, and Full Compliance

The participation constraint (PC) requires that the emission standard $\bar{z}_c$ is below the unabated emission intensity z_{jc} . The incentive constraint (IC) requires that the penalty τ_c exceeds the abatement cost factor b_{jc} . Positive abatement effort will occur in the south-east quadrant. Full compliance occurs if the penalty τ_c is sufficiently large, or the emission standard $\bar{z}_c$ are sufficiently low.



exceeds the emission standard $\bar{z}_c$. When this participation constraint is fulfilled, the first order conditions for a profit maximum is

$$\partial \pi_j^R / \partial \theta_{jc} = -q_j z_{jc} + \tau_c e^{-\theta_{jc}/b_{jc}} / b_{jc} \leq 0 \quad \perp \quad \theta_{jc} \geq 0 \quad (10)$$

where $\perp$ denotes complementary slackness. Consequently, the optimum abatement activity is

$$\theta_{jc} = b_{jc} \ln[\tau_c/b_{jc}] \quad \text{if} \quad \tau_c > b_{jc} \quad (11)$$

and $\theta_{jc} = 0$ otherwise. The condition $\tau_c > b_{jc}$ is the incentive constraint (IC) that implies that the expected penalty τ_c must exceed the abatement cost factor b_{jc} .¹¹ The firm may not always fully comply with a standard $\bar{z}_c$, and a firm will not increase θ_{jc} beyond what is necessary to achieve $z_{jc} \exp(-\theta_{jc}/b_{jc}) = \bar{z}_c$. This implies that $\theta_{jc} \leq b_{jc} \ln[z_{jc}/\bar{z}_c]$, with equality in the case of full compliance (FC). Combining this result with (11) determines that partial compliance occurs when $z_{jc} > \bar{z}_c \tau_c / b_{jc}$, i.e., when the initial pollution intensity is high. Conversely, full compliance becomes more likely when the the penalty b_{jc} is set high, or when the emission standard $\bar{z}_c$ is set high (and is thus easy to reach).

Denote constraint bindings as PC: $\bar{z}_c < z_{jc}$, IC: $\tau_c > b_{jc}$, and FC: $\bar{z}_c \geq z_{jc} b_{jc} / \tau_c$; and let $\overline{\text{PC}}$, $\overline{\text{IC}}$, and $\overline{\text{FC}}$ denote their negations. Combining the PC, IC, and FC constraints provides the basis for calculating firm j 's emissions of pollutant c :

$$Z_{jc} = \begin{cases} q_j z_{jc} & \text{if } \overline{\text{PC}} \vee \overline{\text{IC}} \\ q_j z_{jc} b_{jc} / \tau_c & \text{if } \text{PC} \wedge \text{IC} \wedge \overline{\text{FC}} \\ q_j \bar{z}_c & \text{if } \text{PC} \wedge \text{IC} \wedge \text{FC} \end{cases} \quad (12)$$

¹¹The terms 'participation constraint' and 'incentive constraint' are borrowed from the principal-agent literature. Their meaning is changed, however. Here, participation and incentive constraints are governed by technological factors.

Defining the optimal abatement intensity as

$$\phi_{jc} \equiv \max \left\{ 1, \min \left\{ \frac{z_{jc}}{\bar{z}_c}, \frac{\tau_c}{b_{jc}} \right\} \right\} \geq 1 \quad (13)$$

allows equation (12) to be written as

$$Z_{jc} = \frac{q_j z_{jc}}{\phi_{jc}} \quad (14)$$

Under regulation, a firm's environmental unit expenditure A_{jc} is the sum of abatement costs and penalties:

$$A_{jc} = q_j \left\{ \begin{array}{ll} 0 & \text{if } \overline{PC} \\ \tau_c(z_{jc} - \bar{z}_c) & \text{if } PC \wedge \overline{IC} \\ [z_{jc}b_{jc}[1 + \ln(\tau_c/b_{jc})] - \bar{z}_c\tau_c] & \text{if } PC \wedge IC \wedge \overline{FC} \\ z_{jc}b_{jc} \ln(z_{jc}/\bar{z}_c) & \text{if } PC \wedge IC \wedge FC \end{array} \right\} \quad (15)$$

Note that these higher penalties and higher emission standards lower the government's revenue because higher penalties reduce emissions. Using (13), equation (15) can also be written as

$$A_{jc} = q_j \left[z_{jc}b_{jc} \ln \phi_{jc} + \tau_c \max \left\{ 0, \frac{z_{jc}}{\phi_{jc}} - \bar{z}_c \right\} \right] \quad (16)$$

The firm's total environmental costs are $A_j \equiv \sum_c A_{jc}$.¹²

There are four sources of firm heterogeneity in this model: variation in the unabated emission intensity z_{jc} ; variation in abatement cost b_{jc} ; variation in firm size q_j ; and variation in the surrounding population $n_j(r)$. Heterogeneity in z_{jc} and b_{jc} for a particular pollutant may be rooted in, for example, different vintages of technology.

2.4 Symmetric Solution

To illustrate key properties of the model, consider the symmetric case where $\forall j : b_{jc} = b_c$ and $\forall j : z_{jc} = z_c$. The symmetric case has a particularly simple solution. The policy maker will adopt a regulatory menu that will satisfy the PC and IC if the environmental exposure of the population is sufficiently large. When this is the case, the best strategy is to set a menu that ensures full compliance.¹³

¹²Regulation may also have the result of forcing firms to exit when either τ is set too high and/or $\bar{z}$ too low for one or more pollutant. Exit will be induced when $p - c < A_j/q_j$, in which case emissions fall to zero.

¹³This is easily proved. Suppose a menu was set such that for a given $\tau_c > b_c$, the $\bar{z}_c < z_c b_c / \tau_c$ assured only partial compliance. Then the policy maker would find it beneficial to increase $\bar{z}_c$ in order to reduce penalty payments, and would continue raising $\bar{z}_c$ until $\bar{z}_c = z_c b_c / \tau_c$.

The optimal regulatory policy in the symmetric case is very simple. For notational convenience, let $Q \equiv \sum_j q_j$ and $M \equiv \sum_j n_j q_j$. Further let

$$\psi_c \equiv \gamma \kappa_c M / Q \quad (17)$$

denote an exposure measure for pollutant c , weighted by the policy intensity parameter γ . Then $\partial \pi_j^R / \partial \bar{z}_c = 0$ can be solved for the optimal emission standard $\bar{z}_c = b_c z_c / \psi_c < z_c$. This emission standard will be set only when $\psi_c > b_c$; otherwise the optimal menu is no regulation at all. The condition $\psi_c > b_c$ implies that the health cost from emissions exceed the abatement unit cost b_c . Which penalty τ_c must be set? The FC requirement implies $\tau_c = b_c z_c / \bar{z}_c$, and thus $\tau_c = \psi_c > b_c$ supports the choice of optimal $\bar{z}_c$.¹⁴

2.5 Firm Heterogeneity and the Abatement Ladder

Given that there are four sources of firm heterogeneity (size, unabated pollution intensity, abatement cost factor, and population at risk), all of which may be expressed as empirical distributions, it is difficult to find general algebraic expressions for the optimal policies $(\bar{z}_c, \tau_c)$ even in the case where there are only two firms. As optimal intervention design is not the central point of this paper, the general case is left for further study. However, there are some special asymmetric cases that lend themselves to simple yet insightful analysis.

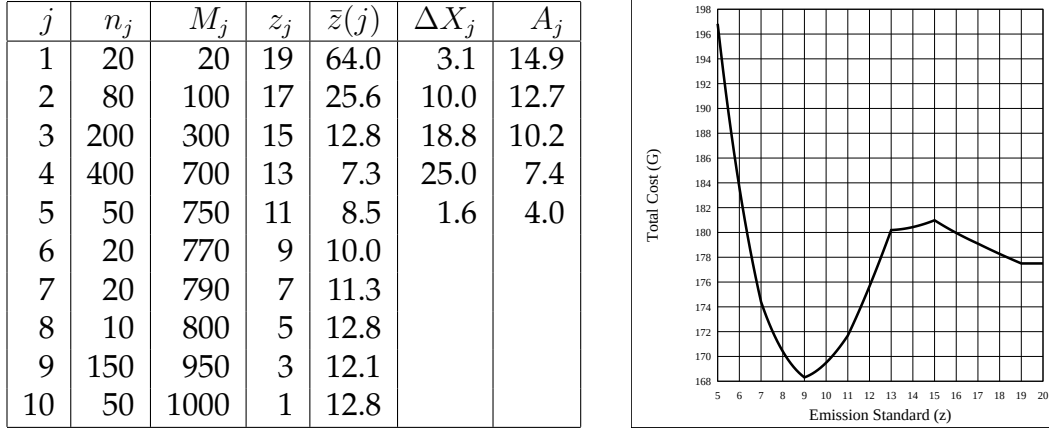
First consider the case where firms' unabated pollution intensity and abatement cost factors are all inversely related such that $\forall j : b_{jc} z_{jc} = B_c$. Then firms with high unabated pollution intensity face the lowest unit abatement cost, and in figure 2 all firms are to be found on the inverse schedule that marks the FC constraint boundary. Put another way, the "low-hanging fruit" for abatement are found in the most polluting firms. Because all firms are situated on the full compliance boundary, the participation and incentive constraints coincide, which further simplifies treatment of this special case.

When all firms engage in some abatement, the optimum $(\bar{z}_c, \tau_c)$ is exactly the same as in the symmetric case: $\tau_c = \psi_c$ and $z_c = B_c / \psi_c$. Ordering firms such that $z_{c1} > z_{c2} > \dots > z_{cJ}$ (or equivalently, $b_{c1} < b_{c2} < \dots < b_{cJ}$) implies that for a given $\bar{z}_c$ either no firms abate ($\bar{z}_c \geq z_{c1}$), some firms abate ($z_{cj} > \bar{z}_c \geq z_{c,j+1}$), or all firms abate ($z_{cJ} > \bar{z}_c$). This ranking constitutes an *abatement ladder* on which a firm's rung is determined by the interaction of the four sources of firm heterogeneity. The boundary between abaters and non-abaters is determined by a measure that is cumu-

¹⁴Of course, $\tau_c = \psi_c$ would also be the optimal solution in the case of a simple pollution tax with $\bar{z}_c = 0$. Without the emission standard $\bar{z}_c$, firms incur a greater total emission cost because the emission standard caps the tax payment. It can be shown that with no emission standard, $G^*(\tau_c = \psi_c, \bar{z}_c = 0) = Q \sum_c z_c b_c [2 + \ln(\psi_c / b_c)]$, which is necessarily larger than the case with an optimal emission standard: $G^*(\tau_c = \psi_c, \bar{z}_c = b_c z_c / \psi_c) = Q \sum_c z_c b_c [1 + \ln(\psi_c / b_c)]$.

lative over the environmental exposure of firms below on the abatement ladder. To determine this border, let $M_j \equiv \sum_k^j n_k q_k$, let $Q_j \equiv \sum_k^j q_k$, and let $\psi_c(j) = \gamma \kappa_c M_j / Q_j$. When j firms participate in pollution abatement, the optimal emission standard should be $\bar{z}_c(j) = B_c / \psi_c(j)$. But because it must hold that $z_{cj} > \bar{z}_c(j) \geq z_{c,j+1}$, emission standard $\bar{z}_c(j)$ may not be feasible. If none of the $\bar{z}_c(j)$ is feasible, the optimal $\bar{z}_c$ is determined by inspecting the cost function at each z_{jc} .

Figure 3: A Numerical Example of Regulation with Firm Heterogeneity



Note: This numerical example was calculated for a single pollutant, inversely related unabated pollution intensity and abatement cost factor ($\forall j : b_j z_j = B = 20$), equal firm size ($\forall j : q_j = 1$), and $\gamma \kappa = 1/64$. Column $\bar{z}(j)$ shows the required emission standard if the top j firms participate in pollution abatement. $\Delta X_j = \gamma \kappa n_j q_j (z_j - \min\{z_j, \bar{z}\})$ shows firm j 's reduction in environmental exposure at the optimal emission standard ($\bar{z} = 9$), and A_j shows the total abatement cost for firm j .

A numerical example, shown in figure 3, illustrates the economic intuition behind the determination of the optimal emission standard and emission tax. This example also reveals that the optimal $\bar{z}_c(j)$ is frequently a particular z_{jc} . The example is for a single pollutant, and thus subscript c is dropped for convenience. The model parameters for ten firms are shown in the table, arranged in descending order of z_j , in equal step sizes from $z_1 = 19$ to $z_{10} = 1$. The figure shows that as the emission standard is toughened ($\bar{z}$ is lowered), total cost increases initially. As seen in the table, the two dirtiest (ie, most pollution intensive) polluters expose only a small part of the population. Since $\bar{z}(j) > z_j$, capturing just these two firms is inefficient. Because the third and fourth firm together expose six times more people than the first and second firm, the emission standard ought to capture these targets. When the top four polluters are captured, the government would like to set the standard at 7.3. But the next dirtiest polluter is at $z_5 = 9$, which would also capture firms 5 and 6. When the sixth firm is captured by $\bar{z}$, its relatively high unit abatement cost start offseting the gains from lower pollution. With six firms abating, the government would indeed prefer to set the standard back at 10.0. Inspecting the total cost curve, it is apparent

that capturing 5 firms and setting $\bar{z} = 9$ is the optimal solution. At that level, each of the five abating firm makes a contribution ΔX_j to reducing environmental exposure, at cost A_j , as shown in the table.

Two further special cases of firm heterogeneity are useful to consider: a simple pollution tax ($\tau_c > 0$, $\bar{z}_c = 0$) and a simple emission standard ($\tau_c = \infty$, $\bar{z}_c > 0$). Defining $M_j \equiv \sum_k^j q_k n_k$, $W_{jc} \equiv \sum_k^j q_k n_k b_{kc} z_{kc}$ and $V_{jc} \equiv \sum_k^j q_k b_{kc} z_{kc}$, the optimal pollution tax for j participants is found as $\tau_c(j) = \gamma \kappa_c W_{jc} / V_{jc}$, and the optimal emission standard with j participants is $\bar{z}_c(j) = V_{jc} / (\gamma \kappa_c M_j)$. Here, k denotes the marginal participating firm, and the ranking of firms is determined by their pollution abatement cost (in the case of the emission tax) and their unabated pollution intensities (in the case of an emission standard). As in the first special case, if none of the $\tau_c(j)$ or $\bar{z}_c(j)$ is feasible (ie., $\tau_c(j) > b_{j+1,c}$ or $\tau_c(j) < b_{jc}$; or $\bar{z}_c(j) > z_{jc}$ or $\bar{z}_c(j) < z_{j+1,c}$) the government cost function must be inspected at each b_{jc} (or z_{jc}) to determine the optimal regulation.

What are the main lessons from studying firm heterogeneity? First, the above analysis highlights the importance of the participation and incentive constraints for observing actual abatement effort. For any given regulatory intervention—emission tax, emission standard, or a combination thereof—there may be a set of active abaters and a set of non-abaters. Which group a particular company falls into depends on its rank on the “abatement ladder,” which is determined in a non-trivial fashion by the interaction of the four sources of firm heterogeneity. Second, the boundary between abaters and non-abaters is determined by—or at the very least highly correlated with—a “mass index” that is cumulative over the population exposure and size of firms at the same level or higher up on the “abatement ladder.” These key insights provide the basis for the empirical analysis below.

2.6 Regulatory Threat

A model of regulatory threat was introduced by Glazer and McMillan (1992) in the context of regulating monopolies. It is useful for contemplating environmental regulatory threat as well. The threat is modelled stochastically: in each period, the government may introduce regulation with probability λ .

Consider firm j 's problem of maximizing its infinite-period present-discounted value V_j , where future periods are discounted at rate ρ . One-period profits under regulation and no regulation are π_j^R and π_j^N , respectively, as defined by equations (9) and (8). In a given pre-regulation period, the firm obtains π_j^N . In the next period it faces either regulation with probability λ , or a continuation of the unregulated regime with probability $1 - \lambda$. When regulation is introduced, it is irreversible, and thus the present-discounted value of the firm is $V_j^R = \pi_j^R / \rho$. In the present period, the firm's

value is determined recursively:

$$\begin{aligned}
V_j &= [\pi_j^N + \lambda V_j^R + (1 - \lambda)V_j] / (1 + \rho) \\
&= [\pi_j^N + \pi_j^R \lambda / \rho] / (\lambda + \rho) \\
&= V_j^R + [\pi_j^N - \pi_j^R] / (\lambda + \rho)
\end{aligned} \tag{18}$$

The second line in equation (18) provides the basis for the derivations below. The third line illustrates nicely that lower regulatory threat (a lower λ) increases the value of the firm, taking the firm's value under regulation as the reference point. The first-order condition for a profit maximum is

$$\begin{aligned}
\frac{dV_j}{d\theta_{jc}} &= \frac{\partial V_j}{\partial \theta_{jc}} + \frac{\partial V_j}{\partial \lambda} \frac{\partial \lambda}{\partial \theta_{jc}} \\
&= \frac{1}{\lambda + \rho} \left[\frac{\partial \pi_j^N}{\partial \theta_{jc}} - \frac{\pi_j^N - \pi_j^R}{\lambda + \rho} \frac{\partial \lambda}{\partial X} \frac{\partial X}{\partial \theta_{jc}} \right] \\
&\leq 0 \quad \perp \quad \theta_{jc} \geq 0
\end{aligned} \tag{19}$$

From (8), (7), and (3), respectively, it follows that

$$\frac{\partial \pi_j^N}{\partial \theta_{jc}} = -q_j z_{jc} \tag{20}$$

$$\frac{\partial \lambda}{\partial X} = (1 + \zeta) \lambda^{\zeta/(1+\zeta)} / X^* \tag{21}$$

$$\frac{\partial X}{\partial \theta_{jc}} = -\kappa_c n_j q_j z_{jc} \exp(-\theta_{jc}/b_{jc}) / b_{jc} \tag{22}$$

Further let $q_j \varphi_j(\theta_j, \bar{z}, \tau) \equiv \pi_j^N(\theta_j) - \pi_j^R(\bar{z}, \tau)$. First-order condition (19) can then be written as¹⁵

$$\theta_{jc} = b_{jc} \ln \left[\frac{(1 + \zeta) \lambda^{\zeta/(1+\zeta)}}{\lambda + \rho} \frac{\kappa_c n_j q_j \varphi_j^*(\bar{z}, \tau)}{b_{jc} X^*} \right] \geq 0 \tag{23}$$

Similar to the case of regulation, the firm faces an incentive constraint for voluntarily abating pollution. Abatement becomes more likely when a firm is large, located in a populated area, production involves toxic substances, and abatement is cheap. The effect of changes in regulatory threat is more complex:

$$\frac{\partial \theta_{jc}}{\partial \lambda} = \frac{b_{jc}(\zeta \rho - \lambda)}{\lambda(\lambda + \rho)(1 + \zeta)} \tag{24}$$

For regulatory threat to be effective and lead to increases in abatement effort θ_{jc} , the above expression must be positive. This implies that the regulation probability λ has an upper bound: $\lambda < \zeta \rho$. This means that regulatory threat is only effective if it is not too large. If it exceeds $\zeta \rho$ and implementation is likely within a short time, firms will find it more profitable to simply wait

¹⁵This solution ignores the second-order repercussion effect through which θ_{jc} affects ϕ_j . The function void of the effect of θ_{jc} is denoted as $\varphi_j^*(\tau, \bar{z})$.

for the implementation and benefit from the lower production cost meanwhile. This is perhaps a perverse effect of regulatory threat: it must not be too strong to succeed. In figure 1, governments prefer the flat part of $\Lambda(X)$; they will use regulatory threat only if regulation is sufficiently distant in the future. If firms have very low discount rates, regulatory threat will become ineffective as firms only care about their current-period profits. Note that a higher ζ (an increase in the curvature of $\Lambda(X)$) expands the usefulness of regulatory threat as an instrument. The optimal regulatory threat is achieved when increases in λ lead to no further increases in abatement effort θ_{jc} , and $\partial\theta_{jc}/\partial\lambda$ becomes zero. This implies an optimal threat

$$\lambda = \zeta\rho \quad (25)$$

By how much do firms reduce emissions? A suitable measure is the ratio of emission intensities with and without abatement. Combining the emission equation $Z_{jc} = q_j z_{jc} \exp(-\theta_{jc}/b_{jc})$ with (23) and (25), it follows that

$$\frac{Z_{jc}/q_j}{z_{jc}} = \exp\left[\frac{\theta_{jc}}{b_{jc}}\right] = \left[\frac{b_{jc}X^*}{\kappa_c n_j q_j \varphi_j^*(\bar{z}, \tau)}\right] \cdot \left[\frac{\zeta}{\rho}\right]^{\frac{1}{1+\zeta}} \leq 1. \quad (26)$$

Firms that are large, put greater number of people at risk through their location, or whose production involves toxic substances, will have larger emission intensity reductions because these firms wield greater influence on the probability of implementing regulation. Furthermore, voluntary abatement is also increased by the cost of the threatened regulation $\varphi_j^*(\bar{z}, \tau)$. Thus a firm's abatement effort will critically depend on whether the firm meets the participation and incentive constraints of actual regulation: its rung on the abatement ladder. Firms for which these constraints are not binding will not see any reduction in emissions. Furthermore, increases in the threatened emission tax τ_c or tougher (lower) emission standards $\bar{z}_c$ will tend to lower π_j^R , increase φ_j^* , and thus increase voluntary abatement. Equation (26) provides the basis for the empirical implementation below.

2.7 Strategic Interactions

I complete the theory section with a brief detour. The above analysis started out with the assumption that firms do not act strategically and thus do not take the response of other firms into consideration. If firms attempt to free ride on the abatement effort of other firms, then

$$\frac{dX}{d\theta_{jc}} = \frac{\partial X}{\partial \theta_{jc}} + \sum_{i \neq j} \sum_d \frac{\partial X}{\partial \theta_{id}} \frac{\partial \theta_{id}}{\partial \theta_{jc}} \quad (27)$$

where $\partial\theta_i/\partial\theta_{jc}$ is the *conjectural variation*. Since what matters is total environmental exposure (X), strategic behaviour takes into account abatement effort of other firms, even if it involves different pollutants than c .

If firms act non-strategically, $\partial\theta_{id}/\partial\theta_{jc} = 0$. However, if they free ride, then $\partial\theta_{id}/\partial\theta_{jc} < 0$. Thus, $dX/d\theta_{jc}$ may turn out to be smaller than the non-strategic model outlined in the previous section. In the extreme case, $dX/d\theta_{jc} \rightarrow 0$ if all firms try to fully free ride on each other.

The extent to which pollution-abatement free-riding is a practical problem is an area of active research. Millock and Salanié (2001) provide an assessment of the potential of free-riding, but also show that incentives to free-ride can be diminished when firms cooperate on other activities such as R&D. The extent to which free-riding matters is, in the end, an empirical question that has been rather elusive.

3 Data Issues

3.1 The National Pollutant Release Inventory (NPRI)

This study uses data from Canada's NPRI, which covers 2,686 facilities over the period 1993–99.¹⁶ Facilities (plants) are required to report releases and transfers of 245 different chemical substances. Releases are recorded separately for different release media (air, water, land, underground injection). Furthermore, onsite releases at the plant location are distinguished from off-site transfers to other locations for treatment or storage. Facilities below a certain size (i.e., those with fewer than 10 employees that do not produce or use NPRI substances in quantities greater than 10 tons, and those in certain exempt categories such as educational institutions) are not required to report. Of the roughly 32,000 manufacturing establishments identified by Statistics Canada, less than 8% report to the NPRI.

Not all facilities reporting to the NPRI actually report positive onsite releases. A good many of them report zero releases. In 1993, just 736 facilities reported positive onsite releases. By 1999, this number had increased to 1344. 960 facilities never reported any positive onsite emissions whatsoever (although a small share of them reports positive offsite transfers). Facilities may not always report continuously. There were only 738 facilities that reported positive onsite emissions in both 1994 and 1999, and only 680 of these facilities reported emissions continuously between 1994 and 1999.

Although between 1993 and 1999 facilities were required to report discharges of 245 substances, 73 of which were added to the inventory in 1999,¹⁷ reports were received for only 192 substances, with an average between 3 and 4 substance reports per facility each year.¹⁸ NPRI focuses on

¹⁶NPRI data are accessible on the web at <http://www.ec.gc.ca/pdb/npri>. Database queries for individual facilities or pollutants can be carried out at <http://www.npri-inrp.com/queryform.cfm>.

¹⁷The number of substances was increased again in 2000 to include highly toxic "micropollutants" that are released in small quantities, with a lower reporting threshold applying to those pollutants.

¹⁸In 1999 about one third of all companies report only a single pollutant, about half

“toxic” substances, and thus does not include many traditional pollutants, such as greenhouse gases, particulates released to air, or suspended solids discharged to water. These excluded substances undoubtedly present significant risks to health and the environment, and thus the information in the NPRI does not tell the full story.¹⁹

In Harrison and Antweiler (2001) we have documented extensively trends in onsite pollution releases and offsite transfers. It is worthwhile highlighting some of the key findings from this study. Between 1993 and 1999, onsite emissions dropped have dropped by 27% by weight, equivalent to an 11% reduction after adjustment for toxicity (see next section). At the same time, offsite transfers decreased at most by 9% by weight, but increased by 127% after adjustment for toxicity. We found evidence that regulatory interventions in the pulp and paper industry (CSIC 27) led to a substantial reduction of releases to surface water. This industry was the the only industry that faced new discharge regulations at the national level during this period, and it was also subject to extensive reform of regulations and permits at the provincial level in the early 1990s (Harrison, 1996). The empirical analysis thus excludes this industry to prevent regulation being misidentified as regulatory threat. Also excluded is a single facility in Québec, Kronos, that dramatically reduced its releases in response to regulatory enforcement action by both the federal government and the province.²⁰ With these adjustments for regulatory intervention, total onsite releases increased by 4% by weight, and decreased 13% in toxicity-adjusted terms. In contrast, offsite transfers decreased 8% in weight but increased 127% in toxicity-adjusted terms. This implies that the average toxicity of onsite releases has decreased, while it has significantly increased for offsite transfers.²¹

3.2 Toxicity Adjustments

Following the example of Hettige et al. (1992) and Horvath et al. (1995), and more recently Shapiro (1999), it is necessary to take into account the varying toxicity of NPRI substances. Like Shapiro (1999), I am using the EPA Chronic Human Health Indicator (CHHI) for the purpose of aggregating pollutants.²² An important advantage of the CHHI data is that EPA has de-

report three or fewer pollutants, and less than 9% report ten or more pollutants. The largest number of pollutants reported by any firm is 55.

¹⁹Environment Canada has announced that the NPRI release for the 2001 data will integrate the criteria air contaminants.

²⁰Further excluded are two disputed measurements in Ontario in 1999.

²¹These figures are not adjusted for the larger number of reporting facilities in 1999 compared to 1993. The qualitative result remains unchanged, however, when we focus only on continuous reporters in 1993 and 1999; see Harrison and Antweiler (2001).

²²The earlier studies by Hettige et al. (1992) and Horvath et al. (1995) rely on threshold limit values (TLVs) adopted by the American Conference of Governmental Industrial Hygienists, but this approach has several limitations. TLVs are designed to reflect the toxicity

veloped separate toxicity scores for each chemical depending on whether exposure occurs via oral intake or inhalation. Thus, inhalation scores are applied to releases by air, and oral toxicity scores are applied to all other release streams.²³ CHHI scores are based only on chronic effects and do not consider acute effects of exposure. However, this is arguably more appropriate in light of the low level exposures resulting from most environmental releases. CHHI indicators do not address multiple effects, effects of concurrent exposures to multiple substances, nor environmental impacts other than human health.²⁴ To interpret our numbers, CHHI adjustments are expressed in “methanol units.”

3.3 Population Data

Spatial population data for Canada were obtained from the *Gridded Population of the World* (GPW version 2) project of the Center for International Earth Science Information Network (CIESIN) at Columbia University.²⁵ GPW provides population estimates for 1990 and 1995 for grids of 2.5 arc minutes each for longitude and latitude.²⁶ A 2'.5x2'.5 grid at 49° latitude (e.g., Vancouver) covers approximately 14 km², and it is somewhat taller than wide (4.6 km by 3 km). Figure 4 shows the distribution of Canada's population by areas of different population density. Roughly one third of Canada's population lives in rural or sparsely populated areas (density lower than 250 people per km²), while about one sixth of the population lives in high-density metropolitan areas (density exceeding 2,500 people per km²). Figure 5 confirms that the majority of the Canadian population lives within a few degrees of latitude, suggesting that using grids of equal arc length does not introduce too much of a distortion. Figure 6 displays the population density on a map.

NPRI facility locations, reported as longitudes and latitudes, were matched to corresponding grids. To study the sensitivity of the results with respect to grid size, I also consider larger grids. Centered around the facility location grid, I consider two alternative areas that cover approximately

of airborne substances only, and the same toxicity rankings may not apply to other forms of exposure. Another drawback is that TLVs are developed for occupational settings, where exposures to toxic substances tend to be high relative to ambient environmental exposures. The same toxicity rankings may not hold at lower exposure levels.

²³This assumes that the ultimate source of human exposure from land-based disposal techniques, including landfills, surface land-application, and underground injection will be via surface or ground water contamination.

²⁴CHHI coverage of NPRI substances is incomplete, but over 90% of the raw weight of pollutants can be accounted for. One potential bias is with respect to nitrate; the CHHI does not provide either oral or inhalation scores.

²⁵<http://sedac.ciesin.org/plue/gpw/index.html>

²⁶Currently, the GPW project only provides data points for 1990 and 1995. The release of GPW data points for 2000 is planned, but in the Canadian case is based on the 2001 census data which is only now becoming available.

Figure 4: Population Density Distribution for Canada

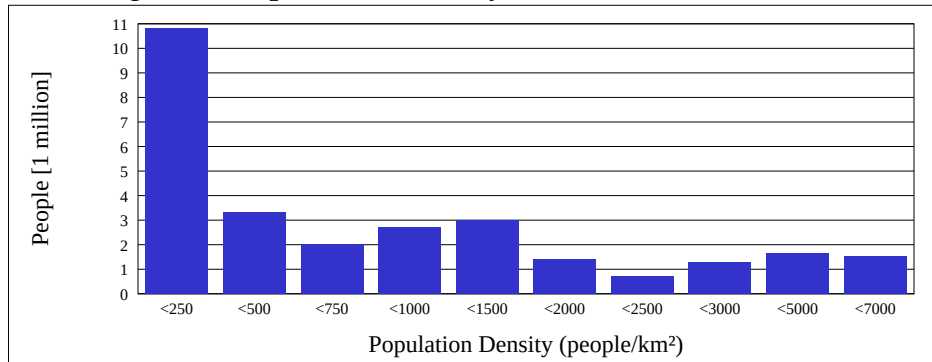


Figure 5: Canadian Population Distribution by Latitude

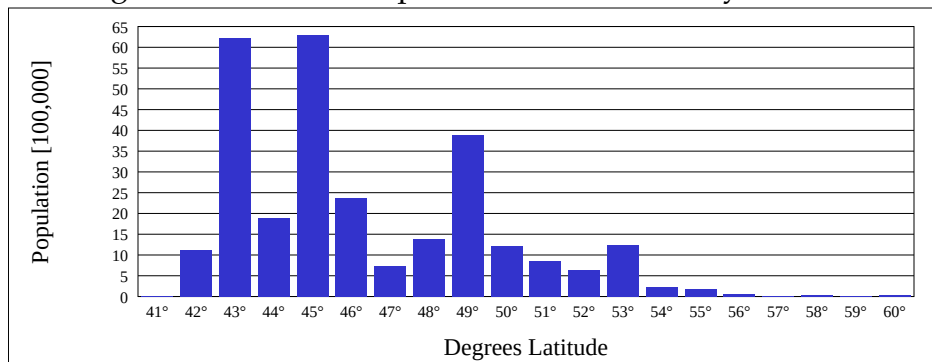


Figure 6: Population Density Map

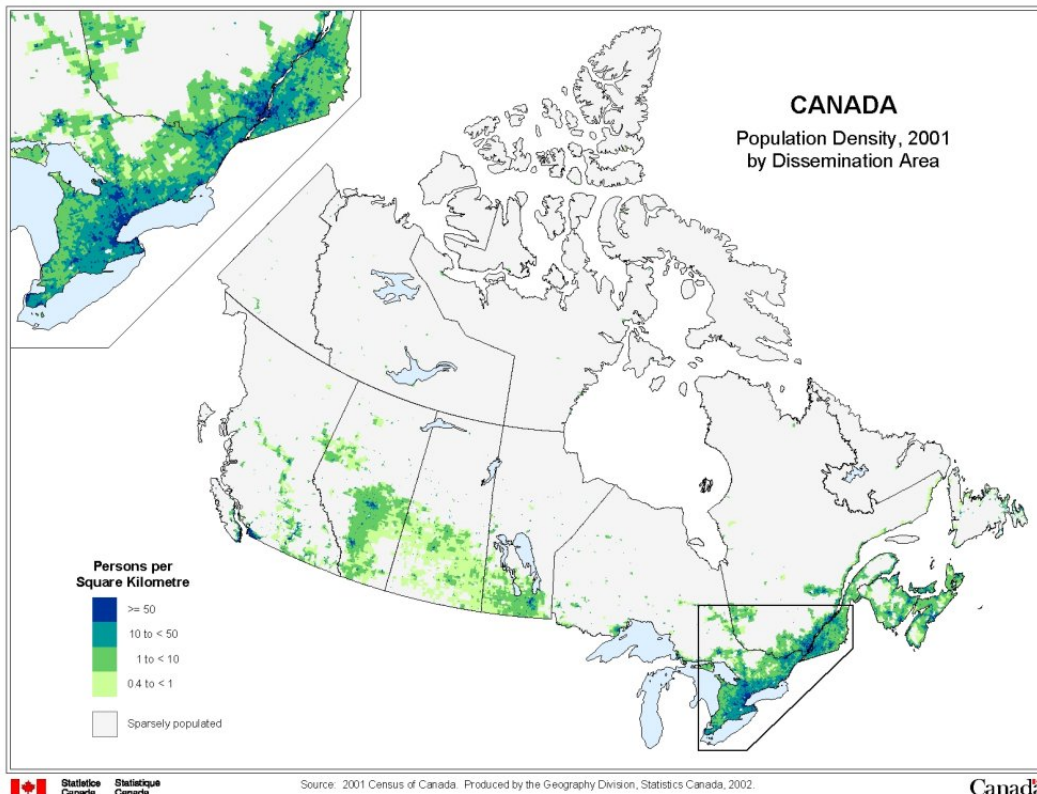
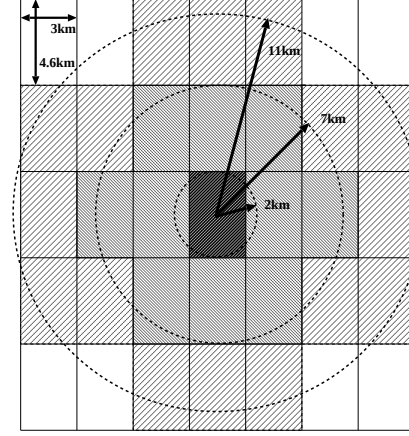


Figure 7: Grid Coverage

Each grid covers an area of 2.5 by 2.5 arc minutes of longitude and latitude. At 49° latitude, this corresponds to an area 3 km wide and 4.6 km high. Aggregation of grids around the center are indicated through different shadings. The three areas considered cover approximately 14 km², 150 km², and 375 km².



150 km² and 375 km². Figure 7 shows how these virtual grids are assembled from 11 and 27 base grids.

4 Environmental Exposure

Central to the analysis in this paper is the calculation of a measure of *environmental exposure*. For a given chemical c , economy-wide environmental exposure in period t is defined as

$$X_{ct} = \kappa_c \sum_j n_{jt} Z_{jct} \quad (28)$$

Table 7 provides a ranking of environmental exposure for the top 32 pollutants, based on the average X_{ct} across 1993-99. The table also includes summary statistics for each pollutant: its EPA-CHHI weights for oral intake and inhalation; the decomposition of onsite emissions into the four principle emission streams (air, water, subsoil, land); the decomposition of total emissions into onsite releases and offsite transfers; and the total onsite emissions in tons per year. Chemical compounds are identified by their name and CAS number, and a map of years in which the chemicals were reported to the NPRI. The exposure measure is reported in logarithmic form due to its wide span. The top pollutant (lead compounds, $\ln X_c = 20.03$) has an exposure measure 20 times that of the second-ranked pollutant (sulfuric acid, $\ln X_c = 17.03$), and 400 times that of the 12th-ranked pollutant (benzene, $\ln X_c = 14.02$).

An interesting question is which contributing factor (population weights, toxicity weights, or emission size) explains the variance in the environmental exposure measure. Using rank correlations, emission size had the lowest and statistically insignificant rank correlation with environmental exposure. The toxicity weights had a rank correlation of 0.80, and the population weights had a rank correlation of 0.60, both statistically significant. Toxicity-adjusted emissions had a rank correlation of 0.96, identifying the toxicity adjustments as the primary source of variation in environmental exposure.

Under section 11 of the 1999 revision of the Canadian Environmental Protection Act (CEPA), Environment Canada can identify substances as toxic and place them on what is known as CEPA Schedule 1. This is in addition to the substances identified as carcinogens in the original CEPA. These substances are identified as “Sch.1” and “Toxic” in tables 7 and table 1. The main distinction between the two groups is that the Schedule 1 substances underwent a review process determining toxicity, while the “Toxic” substances were already deemed “toxic” when CEPA was introduced. Substances identified as toxic and are added to Schedule 1 may become subject to regulation at a later point. As of May 2002 there were 52 substances, but not all are covered by the the NPRI.²⁷ Not all of the substances are among the pollutants with the largest environmental exposure.

It is possible to construct a firm-specific environmental exposure index analogous to (28):

$$X_{jt} = n_{jt} \sum_c \kappa_c Z_{jct} \quad (29)$$

This measure can be used to identify the environmental impact of individual plants. The top thirty facilities are identified in table 2. In addition to the exposure measure $\ln X_{jt}$, the table also shows toxicity-adjusted emissions $\sum_c \kappa_c Z_{jct}$. Steel mills and other metal processing factories are particularly prominent among them. Furthermore, many of the facilities can be found in the heavily-populated Hamilton-Toronto-Montréal corridor. Three alternative rankings are provided in columns a, b and c, based on the three different population area measures introduced in the previous section. The general tendency of these alternative measures is to give greater weight to the facilities in heavily-populated areas. The overall impression is, however, that the ranking is quite consistent across the three specifications.

5 Empirical Implementation

The ideal dependent variable, if it was observable, would be the ratio of abated to unabated emission intensities $(Z/q)/z$, or in logarithmic form $\ln(Z) - \ln(q) - \ln(z)$. However, without data on the abatement effort of individual facilities, it is not possible to observe z . Observable are emissions Z_{jct} of pollutant c by facility j in year t . Also observed are the number of employees at a particular facility, which will be used as a crude proxy for a facility’s output q_{jt} . (Unfortunately, the NPRI database does not reveal a facility’s output either by value or volume.) Because unabated emission intensities are unobserved, the z_{jc} ’s may be thought of as time-invariant fixed effects.

It would be natural to choose $\ln Z_{jct}$ as the dependent variable and relegate $\ln q_{jt}$ and $\ln z_{jc}$ to the right-hand side of the estimating equation; q

²⁷http://www.ec.gc.ca/CEPARegistry/subs_list/Toxicupdate.cfm

because it is a proxy that will tend to introduce mismeasurement problems, and z because it is an unobserved fixed effect. The key regressor on the right-hand side would be the environmental exposure X_{jct} , also suitably expressed in logarithmic form. This set-up is, in its simplest form,

$$\ln Z_{jct} = \ln z_{jc} + \beta \cdot \ln q_{jct} - \gamma \cdot \ln X_{jct} \quad (30)$$

where β and γ are estimable parameters. Parameter β is the scale elasticity of emissions, and parameter γ captures the effect of regulatory threat and should yield a significant positive estimate if regulatory threat is effective.

There are two immediately identifiable econometric complications—plus an economic identification problem. First, treating z_{jc} as unobservable requires fixed effects for facilities and pollutants; let α_{jc} denote these fixed effects. This leaves just the time dimension—and no cross-sectional dimensions—to identify the effect of environmental exposure (X_{jct}) on emissions (Z_{jct}). Since the NPRI has only seven years of usable data, and many facilities report only in some but not all years, the power of the econometric tests is severely diminished.

Second, because $\ln X_{jct}$ can be decomposed into the sum of $\ln Z_{jct}$, $\ln \kappa_c$ and $\ln n_{jt}$, the dependent variable appears on the right-hand side of the equation. This requires consolidating the $\ln Z_{jct}$ terms in (30) on the left-hand side and redefining $\tilde{\alpha}_{jc} \equiv \alpha_{jc}/(1+\gamma)$, $\tilde{\beta} \equiv \beta/(1+\gamma)$, and $\tilde{\gamma} \equiv \gamma/(1+\gamma)$, respectively.²⁸ The estimate of $\tilde{\gamma}$ can then be used as a predictor of the original $\gamma = \tilde{\gamma}/(1-\tilde{\gamma})$ through application of the Delta method; see Greene (2003, p. 70).

Third, can one really attribute a positive γ estimate to federal regulatory threat? Could it perhaps also capture the effect of local regulation and the location choice of firms? It may simply be the case that firms with lower pollution intensity choose urban areas, and firms with high pollution intensity choose uninhabited areas. Then environmental exposure and emission intensities will be negatively correlated. Alas, this does capture a form of fear of regulation. If (manufacturing) firms avoid urban areas, it is precisely because of the fear that it is more likely that they will face environmental regulation. Of course, high-intensity polluters may also be subject to local regulation such as zoning restrictions. This suggests that the estimate of γ will tend to overstate the effect of regulatory threat because it may capture the effect of local regulations and zoning. Hence, further empirical instruments are needed to disentangle the effect of regulatory threat. I devise two such instruments. The first instrument is suggested by the theoretical

²⁸Theory suggests that γ applies to $\ln \kappa_c + \ln n_{jc}$. One cannot estimate partial effects on $\ln \kappa_c$ and $\ln n_{jc}$ because κ_c has (theoretically) no time variation and would be captured by the fixed effects. In practice, κ_c may have minimal time variation. Facilities may vary emission streams over time. Because of different oral and inhalation toxicities, changes in the composition of emission streams may affect the average toxicity of a particular pollutant at a given facility.

model: conditioning firms' abatement effort on their environmental exposure ranking relative to other firms. The second instrument is an empirical identification strategy: investigating rates of change in emissions rather than emission levels.

Turning first to emission level regressions, the following estimating equations will be used: a parametric nonlinear equation

$$\ln Z_{jct} = \tilde{\alpha}_{jc} + \tilde{\beta} \cdot \ln q_{jct} - \tilde{\gamma} \cdot \ln(\kappa_c n_{jt}) \cdot \xi_{jct}^\delta + \epsilon_{jct} \quad (31)$$

and its non-parametric cousin

$$\ln Z_{jct} = \tilde{\alpha}_{jc} + \tilde{\beta} \cdot \ln q_{jct} - \tilde{\gamma} \cdot \ln(\kappa_c n_{jt}) \cdot I(\xi_{jct} > \bar{\xi}) + \epsilon_{jct} \quad (32)$$

Here $\tilde{\alpha}_{jc}$ are fixed effects that capture the variation in z_{jc} , and $I(\cdot)$ again denotes the indicator function. We estimate both equations by nonlinear fixed-effects maximum-likelihood.²⁹

The discussion in the theory section of this paper has emphasized the importance of conditioning changes in emissions on the application of participation and incentive constraints: a firm's rung on the abatement ladder. Because only pollution intensities are observed in practice, one can only make inferences about the participation constraint (PC). Unfortunately, the NPRI does not shed any light on abatement cost factors, and thus it is not possible to capture the effect of the incentive constraint (IC) except in the case where the two are highly correlated.³⁰ The PC is tied to a facility's emission intensity with respect to that of other facilities, or short, its emission intensity rank. The highest emitters face the greatest likelihood of a binding PC, and vice versa. The theoretical analysis has shown that it is not the rank alone that matters, but also the environmental exposure enmassed by the facilities with higher pollution intensity. I use this insight to construct the abatement ladder index

$$\xi_{jct} = 1 - \frac{\sum_k X_{kct} I(z_{kct} > z_{jct})}{\sum_k X_{kct}} \in]0, 1] \quad (33)$$

where $I(\cdot)$ denotes the indicator function. A facility with $\xi_{jct} = 1$ is the worst polluter, as no other facilities have a higher z_{cjt} than facility f . Conversely,

²⁹The fixed effects are nuisance parameters. Given that there are several thousand such parameters, it would be grossly inefficient to estimate them by brute force. Fortunately, fixed effects in nonlinear models can be concentrated out of the likelihood function easily. The model can be viewed as a nonlinear panel regression $y_{it} = \alpha_i + f(x_{it}, b) + \epsilon_{it}$ with groups i and time t . Let $u_{it} \equiv y_{it} - f(x_{it}, b)$, and purge the effect of α_i by using the transformation $e_{it} \equiv u_{it} - (1/T) \sum_s u_{is}$. Then the concentrated log-likelihood function is simply $\mathcal{L} = -[\ln(\pi\sigma^2) + \sum_i \sum_t e_{it}^2/\sigma^2]/2$. While equation (32) is entirely linear and could also be estimated by conventional fixed effects, equation (31) is nonlinear in ξ . In order to use analytic gradients in the estimation procedure, note that $\partial f/\partial \delta = -\gamma \ln(\kappa n) \xi^\delta \ln(\xi)$. Because of the $\ln(\xi)$ term, it must hold that $\xi > 0$.

³⁰There is good reason to believe that there is an inverse correlation between z_{jc} and b_{jc} if pollution abatement technology is subject to vintage effects.

ξ_{jct} is approaching zero (but will not be exactly zero) for the cleanest facility. ξ_{jct} can be interpreted as the likelihood that facility j will be affected by the implementation of regulation.

How does the abatement ladder index ξ_{jct} condition the effect of environmental exposure on emissions? If participation constraints are binding for some firms and not binding for others, then there exists a $0 < \bar{\xi} < 1$ that separates the two groups. For the firms captured by the constraint $\xi_{jct} > \bar{\xi}$. Equation (32) captures this by multiplying $\ln(\kappa_c n_{jt})$ with the indicator variable $I(\xi_{jct} > \bar{\xi})$. I perform a simple grid search by using $\bar{\xi} \in \{0, 0.25, 0.5, 0.75\}$ in the regressions. However, the effect of ξ may be nonlinear. To capture this possibility, in estimating equation (31) I condition the effect of environmental exposure on ξ_{jct}^δ . Consistency with the theory requires that $\delta > 0$. When $\delta \rightarrow 0$, ξ_{jct}^δ becomes increasingly concave. Small values of δ filter out the observations with low ξ 's, similar to a low $\bar{\xi}$ in equation (31). Conversely, as $\delta \rightarrow \infty$, all but the observations with the highest ξ are filtered out as the curvature of ξ_{jct}^δ becomes increasingly convex. When the estimated δ is negative, clean facilities react stronger to their environmental exposure when ξ is small. A negative δ thus contradicts the theory, which requires that the most exposed facilities reduce emissions the most.

Estimating equations (31) and (32) remain unsatisfactory. The empirical burden imposed on the functional form is severe, and the effect of regulatory threat is identified entirely through time variation and the conditioning effect imposed by the participation constraints. Furthermore, γ is assumed to be identical across pollutants. Because of these shortcomings I look for alternative empirical implementations.

One possibility is to relax the assumption that z_{jc} is facility-specific, and allow for mere industry effects. This is the alternative that pursued in an earlier paper (Harrison and Antweiler, 2001). It provided the basis for our use of facility aggregates where emissions are adjusted for toxicity and summed over all pollutants. While this approach is appealing from a practical point of view, it does not provide for a differentiated treatment of individual pollutants.

Here I pursue a different empirical strategy to identify the effect of regulatory threat. If in response to regulatory threat facilities reduce emissions gradually over time rather than abruptly, the effect of regulatory threat should be observable over several time periods. The first number of years of the NPRI are a particularly suitable time period as facilities are learning, for the first time, about their environmental exposure relative to that of other firms. The first decade of the NPRI is thus very likely characterized by an adjustment process during which firms are potentially responding to regulatory threat.

Concretely, an alternative empirical implementation can be derived by time-differencing estimating equation (30). The dependent variable becomes the rate of change in observed emissions $\Delta \ln Z_{jct} \equiv \ln Z_{jct} - \ln Z_{jct-1}$. Time-differencing eliminates the fixed effects $\alpha_{jc} = \ln z_{jc}$ and further in-

roduces the rate of change in employment $\Delta \ln q_{jt} \equiv \ln q_{jt} - \ln q_{j,t-1}$. Using the rate of change $\Delta \ln Z_{jct}$ as the dependent variable also provides for cross-sectional comparability in our estimating equation; it also greatly reduces the potential for heteroskedasticity. Furthermore, as emission levels are highly auto-correlated, $\Delta \ln Z_{jct}$ models a stationary I(1) process.

The identification strategy is to include both a time-differenced and a previous-period level term corresponding to the environmental exposure measure. The time-differenced expression $\Delta \ln X_{jct} \equiv \ln X_{jct} - \ln X_{jct,t-1}$ captures the impact of environmental exposure in steady-state. Since increases in environmental exposure should reduce emissions, the corresponding estimator η is introduced with a minus sign. The previous-period level term $\ln X_{jct,t-1}$ captures the transition dynamic to the steady state. In the regressions involving time-differencing we can use $\ln X_{jct,t-1}$ directly because it is lagged. However, similar to estimating equations (31) and (32), it is necessary to consolidate $\Delta \ln X_{jct}$ by adding $\eta \Delta Z_{jct}$ on both sides of the equation. This redefines $\tilde{\alpha} = \alpha/(1+\eta)$, $\tilde{\beta} = \beta/(1+\eta)$, $\tilde{\gamma} = \gamma/(1+\eta)$, and $\tilde{\eta} = \eta/(1+\eta)$.

I am now able to set up a second pair of estimating equations. Again there is a fully parametric and a non-parametric version. A possible time trend is captured by an intercept term $\tilde{\alpha}$. The fully parametric estimating equation is

$$\Delta \ln Z_{jct} = \tilde{\alpha} + \tilde{\beta} \cdot \Delta \ln q_{jct} - \tilde{\gamma} \cdot \ln X_{jct,t-1} \cdot \xi_{jct,t-1}^\delta - \tilde{\eta} \cdot \Delta \ln(\kappa_c n_{jt}) + \epsilon_{jct} \quad (34)$$

and its non-parametric cousin is

$$\Delta \ln Z_{jct} = \tilde{\alpha} + \tilde{\beta} \cdot \Delta \ln q_{jct} - \tilde{\gamma} \cdot \ln X_{jct,t-1} \cdot I(\xi_{jct,t-1} > \bar{\xi}) - \tilde{\eta} \cdot \Delta \ln(\kappa_c n_{jt}) + \epsilon_{jct} \quad (35)$$

These two estimating equations provide greater flexibility than their counterparts (31) and (32). Because there are no pervasive fixed effects to worry about, one can estimate (34) and (35) pollutant by pollutant, in addition to estimating a pooled version involving all pollutants.

Lastly, it is interesting to determine the relative contributions of emission level Z_{jct} , toxicity κ_c , and population exposure n_{jt} . I therefore also estimate a version that decomposes $\gamma \cdot \ln X_{jct,t-1}$ into $\gamma_1 \cdot \ln Z_{jct,t-1} + \gamma_2 \cdot \ln \kappa_c + \gamma_3 \cdot \ln n_{j,t-1}$.

6 Results

6.1 Emission Levels

Table 3 presents evidence from emission level regressions based on estimating equations (31) and (32). Estimates for the nonparametric specification are shown in columns (A) through (D), where the $\bar{\xi}$ cutoffs were chosen as 0, 0.25, 0.5, and 0.75, respectively. Turning to the estimate $\gamma = \tilde{\gamma}/(1 - \tilde{\gamma})$ that corresponds to the environmental exposure effect, only the estimate in column (A) is positive and consistent with the theory's prediction. It indicates

an elasticity of 0.436: a 10% increase in environmental exposure is associated with a 4.4% reduction in emissions. However, the estimated elasticities for the higher cutoff levels for $\bar{\xi}$ are all negative and significant, which suggests that the most pollution-intensive facilities actually tend to have higher emissions when they are environmentally exposed. The parametric estimate of δ for the abatement conditioning effect is shown in column (E). Here, the effect of environmental exposure is again consistent with the theory (the estimated γ is 0.46), but the abatement ladder effect (the estimated δ exponent) is negative. Lastly, the scale elasticities β are estimated in the neighbourhood of 0.2—that is, a 10% increase in employment (the plant size proxy) leads to a 2% increase in emissions.

The results from the emission level regressions cast a dark shadow on the validity of the regulatory threat hypothesis. In particular, the negative δ estimate raises numerous possibilities: the theory may be flawed or incomplete; there is no evidence of regulatory threat; the empirical implementation is incorrect; or another mechanism than regulatory threat is determining the outcome. To address potential problems with the empirical implementation, the case for or against the effectiveness of green regulatory threat is further investigated through the alternative empirical specifications (34) and (35) that appear to provide greater econometric robustness.

Why are the cleanest facilities the most responsive to their environmental exposure? This is in fact not inconsistent with the theory if facilities with the lowest pollution intensity are the ones with the lowest unit abatement cost. Thus they would tend to reduce emissions in response to the threat of even low emission taxes. In this case the incentive constraint (which is driven by the level of the threatened emission tax) empirically dominates the participation constraint (which is driven by the level of the threatened emission standard). The empirical implementation was predicated on the assumption that the observable participation constraint ranking is negatively correlated with the unobservable incentive constraint ranking, so that ξ captures both constraints in the same direction. The negative estimate for δ raises the possibility that this assumption is incorrect, and that z and b are perhaps positively correlated.

6.2 Emission Level Changes

The time-differenced estimating equations (34) and (35) provide several econometric advantages over their emission-level counterparts (31) and (32). In addition to dealing effectively with the fixed-effects problem and autocorrelation in the dependent variable, they also provide for a relaxation of the assumption that δ or $\bar{\xi}$ are equal across pollutants. I explore this possibility by estimating versions of estimating equations (34) and (35) where all parameters are pollutant-specific. Results for these pollutant-by-pollutant regressions are shown in table 4 for the parametric implementation based on (35) and in table 5 for the nonparametric implementation based on (34).

Regression results are shown for the top-35 ranked pollutants except those where regressions would have been based on fewer than 20 observations.

The estimates of $\tilde{\eta}$ that correspond to the change in environmental exposure are in most cases positive and significant. There is only one case (sulfuric acid) with a negative and significant estimate that appears inconsistent with the role of environmental exposure in steady state. When significant and positive, the estimates range between 0.2 and 1.5. Because η is not estimated all that precisely, the estimates in the table are shown without the Delta-method transformation that were employed in table 3. Note that an imprecise η (not significantly different from zero) may lead to a sign reversal for $\gamma = \tilde{\gamma}/(1 - \hat{\eta})$ when $\hat{\eta} > 1$.

Turning to the level of environmental exposure, the estimates of $\tilde{\gamma}$ in table 4 are positive whenever they are significant. In those cases they range typically between 0.05 and 0.10. There are two pollutants for which $\tilde{\gamma}$ is noticeably higher: ethylene oxide (0.247) and aluminum fumes or dust (0.171). The estimates of δ are inconclusive. They are never estimated precisely. In the majority of cases they are small and positive (between about 0.05 and 0.15), in some cases they are very large (naphthalene, butadiene), and in a few instances they are very small and negative. Overall, the results for $\tilde{\gamma}$ and δ are broadly consistent with the hypothesis of green regulatory threat for numerous (in particular high-ranked) pollutants. However, the magnitude of the effect is very small. For example, A 10% increase in environmental exposure of manganese compounds only leads to an annual emission reduction of about 0.7%.

The estimates for the scale effect (change in employment) provide a mixed picture. When significant, $\tilde{\beta}$ is positive in five cases and negative in one case. When positive and significant, the estimates are mostly quite plausible, between about 0.4 and 1.5. The scale effect for vinyl chloride (5.4) is suspiciously high, while the estimate for chromium compounds (-1.3) is implausibly negative.

Overall, the two pollutants with the most observations (toluene and methanol) provide a picture that is broadly consistent with the green regulatory threat hypothesis. While weaker for other substances, there are no cases that suggest an outright rejection of the regulatory threat hypothesis.

Table 5 shows the results for the nonparametric method. It focuses on the estimates of $\tilde{\gamma}$ conditional on particular thresholds $\bar{\xi}$; the estimates of the other parameters are suppressed. The first four columns show $\tilde{\gamma}$ estimates for $\bar{\xi} \in \{0, 0.25, 0.5, 0.75\}$, and the last column shows the $\bar{\xi}$ with the highest likelihood. In the first four columns the figures in parentheses are t-statistics, but in the last column they are χ^2 statistics for a log-likelihood ratio test of the unrestricted case $\tilde{\gamma} \neq 0$ against the restricted case $\tilde{\gamma} = 0$. The unrestricted case is always preferred. Similar to the results in table 4, $\tilde{\gamma}$ is always positive when significant, across all four thresholds. In many cases only the $\tilde{\gamma}|\bar{\xi} = 0$ case is significant. However, there are several cases where

a high $\bar{\xi}$ is preferred. This could be taken as evidence of a participation constraint that would trigger abatement only for the plants with the highest pollution intensity. But even in those cases the estimate for $\hat{\gamma}|\xi = 0$ is often higher (albeit less statistically significant). Similar to the results from the emission-level regressions, there is again evidence that the low-intensity polluters seem to react stronger to their environmental exposure than the high-intensity polluters. This may suggest that the exposure rank index ξ may not appropriately capture the order in which plants may commence abatement effort.

Finally, table 6 presents results for regressions across all pollutants, exploiting the cross-sectional information contained in the dataset. Columns (A) and (C) show estimates for the restricted case $\delta = 0$, while columns (B) and (D) show estimates for the unrestricted case. Compared to columns (A) and (B), columns (C) and (D) decompose the environmental exposure X into its three constituent elements. The regressions in (C) and (D) are merely explorative because theory suggests only a joint multiplicative effect, not additively-seperable partial effects. Nevertheless, the results in columns (C) and (D) shed light on the relative importance of the three constituent elements. As could be expected, size and pollution intensity contribute the most to the environmental exposure effect, followed by pollution toxicity and lastly population exposure. The preferred specification is shown in column (B), where it becomes apparent that a 10% increase in environmental exposure leads to annual emission reductions of about 0.5% across pollutants and facilities. Here, the estimate of γ was again obtained through application of the Delta method, i.e., $\gamma = \hat{\gamma}/(1 - \hat{\eta})$. The estimate for the steady-state behaviour of environmental exposure indicates that changes in environmental exposure reduce emissions: a 10% change in X leads to a 3.9% change in Z . Scale effects, as proxied by changes in employment, affect emission levels to a much smaller extent than could be expected. A 10% change in employment translates into a modest 1.1% increase in emission levels.³¹ Lastly, the estimate of δ that captures the effect of the abatement ladder index ξ is positive and significant at around 0.127 in the preferred specification (B). Judging by the negative estimate in column (D), however, this may not be a robust result.

6.3 Extensions

To verify the robustness of the key results in table 6, it is worthwhile looking at two further sets of regressions. In table 7 we show results based on estimating equations (34) and (35) where the dataset only includes pollutants

³¹Quite possibly, employment is slow to adjust in response to changes in output. In this case employment is a poor proxy for capturing scale effects in a sample with not enough longitudinal variation. Furthermore, variations in capital may be a more important determinant than variations in labour input.

that are listed on CEPA Schedule 1 or the original CEPA Toxic list. Presumably, regulatory threat is most pronounced for substances that the government has earmarked for potential regulation. The empirical estimates shown in table 7 do not differ dramatically from the results shown in table 6. The estimate of γ is virtually unchanged; there is no evidence of accelerated emission reductions. However, the estimate of η (the long-term effect) is about 50% larger, which may indicate somewhat greater responsiveness of plants for these types of pollutants.

A second robustness check focuses on the emission stream. Emissions into air are particularly suited for empirical analysis because of the radial attenuation of emissions; other emission streams exhibit more complex attenuation function. While the estimate of $\tilde{\gamma}$ shows little change compared to the baseline (table 6, column B), the estimate of $\tilde{\eta}$ becomes statistically insignificant and exhibits sign instability across specifications. Because $\tilde{\eta}$ is estimated so imprecisely, the Delta-method transformation of $\gamma = \tilde{\gamma}/(1 - \tilde{\eta})$ reduces the significance of γ . If anything, the results should have become statistically stronger by focusing on the emission stream that is more “fitting” for this type of analysis. These weak empirical results certainly do not strengthen the case for the effectiveness of green regulatory threat.

7 Conclusions

Policy makers have turned to the possibility that the threat of environmental regulation may be an effective instrument to reduce harmful emissions by enticing firms into voluntary abatement effort. This paper’s main objective is to identify the effectiveness of *green regulatory threat* based on plant-level data from Canada’s National Pollutant Release Inventory (NPRI).

The theoretical model developed in this paper delivers several important insights into the mechanics of green regulatory threat. First, firms determine their own voluntary abatement effort based on the *environmental exposure* caused by their emissions. Environmental exposure is a measure that combines the effect from (a) the level of emissions as determined by the plant’s size and pollution intensity; (b) the toxicity of the emitted pollutants; and (c) the size of the population at risk.

Second, the abatement response of an individual firm or plant is conditioned on its rung on an *abatement ladder* which marks its position relative to other firms with respect to their (a) unabated emission intensity, and (b) abatement unit cost. When (actual or threatened) regulation involves a combination of emission standards and emission taxes, a plant’s abatement ladder rung is responsible for determining whether or not it will carry out abatement.

Third, regulatory threat is only effective if, perversely, the threat is not too intense. If the threat is too intense (and thus the introduction of regulation is expected to be soon), firms will prefer to hold out and wait for the

regulation, meanwhile benefiting from not incurring abatement costs.

The empirical strategy for identifying the effectiveness of green regulatory threat is predicated on the interaction of a firm's own environmental exposure with an abatement ladder index that captures the cumulative environmental exposure of facilities on lower rungs of the abatement ladder. Due to data limitations, the abatement ladder index used in this paper is based on pollution intensity data only. A measure of environmental exposure is constructed by combining NPRI data with, primarily, pollutant toxicity data from the U.S.-EPA's Chronic Human Health Indicator, and population data from the CIESIN Gridded Population of the World (GPW) database.

The empirical strategy faces several problems and limitations. Regressions based on the level of emissions are subject to fixed effects across facilities and pollutants, acknowledging the lack of data on unabated emission intensities. This leaves just the time dimension for identifying the effect of green regulatory threat. These limitations can be dealt with by also considering time-differenced (rate-of-change) regressions which eliminate the fixed-effects problem as well as address the autocorrelated nature of emission levels. In these regressions the effect of regulatory threat is identified through the assumption that plants adjust to the optimal abatement level over the course of several years after they learn initially about their environmental exposure.

The empirical work paints a picture that is not entirely conclusive. The results from emission level regressions suggest the possibility that regulatory threat may be present, but that the identification assumption (an inverse correlation between abatement unit cost and pollution intensity) may be incorrect. Instead, the "low-hanging fruit" of pollution abatement may perhaps be found at low-intensity polluters. This tends to favor (threatened or actual) emission standards based on "best-of-class" comparisons.

Evidence from the (econometrically superior) time-differenced regressions paints a different picture. There the empirical evidence is broadly consistent with the green regulatory threat hypothesis. Based on the preferred regression (table 6, column B), in response to a 10% increase in environmental exposure, plants reduce emissions only by about 0.5% annually.

The empirical work suffers from numerous shortcomings in the comprehensiveness and quality of data. Faced with these limitations, the empirical results are not altogether conclusive. While the results from different econometric specifications seem to be modestly supportive of the existence of a green regulatory threat effect, the environmental magnitude of the effect is rather small in Canada. Green regulatory threat does not appear to be very effective in reducing emissions.

References

- Antweiler, Werner and Kathryn Harrison** (2003) "Toxic Release Inventories and Green Consumerism: Empirical Evidence from Canada," *Canadian Journal of Economics*, 36 (2), forthcoming.
- Arora, Seema and Shubhashis Gangopadhyay** (1995) "Toward a theoretical model of voluntary overcompliance," *Journal of Economic Behavior and Organization*, 28 (3), 289–309.
- **and Timothy N. Cason** (1995) "An Experiment in Voluntary Environmental Regulation: Participation in EPA's 33/50 Program," *Journal of Environmental Economics and Management*, 28 (3), 271–286.
- **and —** (1999) "Do Community Characteristics Influence Environmental Outcomes? Evidence from the Toxics Release Inventory," *Southern Economic Journal*, 65 (4), 691–716.
- Cadot, Olivier and Bernard Sinclair-Desagné** (1995) "Innovation Under the Threat of Stricter Environmental Standards." Centre interuniversitaire de recherche en analyse des organisations (CIRANO) working paper number 95s-11, Montréal.
- Glazer, Amihai and Henry McMillan** (1992) "Pricing by the Firm Under Regulatory Threat," *Quarterly Journal of Economics*, 107 (3), 1089–1099.
- Greene, William H.** (2003) *Econometric Analysis*, 5th ed., Upper Saddle River, NJ: Prentice Hall.
- Harford, J. and S. Ogura** (1983) "Pollution Taxes and Standards: A Continuum of Quasi-Optimal Solutions," *Journal of Environmental Economics and Management*, 10, 1–17.
- Harrison, Kathryn** (1996) "The Regulator's Dilemma: Regulation of Pulp Mill Effluents in the Canadian Federal State," *Canadian Journal of Political Science*, 29, 469–496.
- Hettige, H. R., E. B. Lucas, and D. Wheeler** (1992) "The Toxic Intensity of Industrial Production: Global Patterns, Trends and Trade Policy," *American Economic Review*, 82 (2), 478–481.
- Horvath, Arpad, Chris T. Hendrickson, Lester B. Lave, Francis C. McMichael, and Tse-Sung Wu** (1995) "Toxic Emissions Indices for Green Design and Inventory," *Environmental Science and Technology*, 29, 86A–90A.
- Lyon, Thomas P. and John W. Maxwell** (2002) "Self-Regulation, Taxation and Public Voluntary Environmental Agreements," *Journal of Public Economics*, forthcoming.

- Maxwell, John W., Thomas P. Lyon, and C. Steven Hackett** (2000) "Self-Regulation and Social Welfare: The Political Economy of Corporate Environmentalism," *Journal of Law and Economics*, 43 (2), 583–617.
- Millock, Katrin and François Salanié** (2001) "Collective Environmental Agreements: An Analysis of the Problems of Free-Riding and Collusion." Centre International de Recherche sur l'Environnement et le Développement, INRA-Toulouse.
- Segerson, Kathleen and Thomas J. Miceli** (1998) "Voluntary Environmental Agreements: Good or Bad News for Environmental Protection?," *Journal of Environmental Economics and Management*, 36 (2), 109–130.
- Shapiro, Marc D.** (1999) "Toxic Exposure and Environmental Equity: Results from the EPA's New Exposure Estimation Model." University of Rochester working paper.
- Stranlund, John K.** (1995) "Public Mechanisms to Support Compliance to an Environmental Norm," *Journal of Environmental Economics and Management*, 28 (2), 205–222.
- Wu, JunJie and Bruce A. Babcock** (1999) "The Relative Efficiency of Voluntary vs Mandatory Environmental Regulations," *Journal of Environmental Economics and Management*, 38 (2), 158–175.

Table 1: Environmental Exposure Ranking

CAS				Name	Reports	J^+	CEPA		EPA-CHHI		Emission Stream Composition [%]						Emission tons/yr.	Exposure $\ln X_c$
							Sch.1	Toxic	oral	inhale	Air	Water	Subsoil	Land	Onsite	Offsite		
1	NA - 08	Lead compounds	●●●●●●	221	yes	no	100	100	41.9	5.5		52.6	27.3	72.7	1,589	20.03		
2	7664-93-9	Sulfuric acid	●●●●●●	309	no	no	$10 \cdot 10^{-6}$	1.4	34.4	65.6		0.1	80.3	19.7	19,862	17.03		
3	10049-04-4	Chlorine dioxide	●●●●●●	45	no	no	9	9	99.9	0.1			100.0		1,330	16.84		
4	NA - 04	Chromium Compounds	●●●●●●	314	no	yes	$170 \cdot 10^{-3}$	86	2.7	3.0		94.3	22.4	77.6	932	16.80		
5	NA - 02	Arsenic compounds	●●●●●●	85	no	yes	3	31	5.8	0.7	90.0	3.5	96.0	4.0	2,362	16.73		
6	NA - 09	Manganese compounds	●●●●●●	312	no	no	$3.6 \cdot 10^{-3}$	36	3.3	11.3		85.4	33.0	67.0	2,706	16.63		
7	111-42-2	Diethanolamine	●●●●●●	84	no	no	$360 \cdot 10^{-3}$	18	14.5	2.1	80.6	2.7	78.9	21.1	763	15.85		
8	NA - 11	Nickel compounds	●●●●●●	227	no	yes	$25 \cdot 10^{-3}$	3.5	68.2	8.4		23.5	56.0	44.0	701	15.61		
9	7723-14-0	Phosphorus (yellow or white)	●●●●●●	20	no	no	25	25	0.6	31.3		68.1	65.8	34.2	18	15.28		
10	NA - 05	Cobalt compounds	●●●●●●	48	no	no		90	26.1	3.2		70.6	83.1	16.9	57	15.01		
11	7647-01-0	Hydrochloric acid	●●●●●●	229	no	no	$90 \cdot 10^{-3}$	$90 \cdot 10^{-3}$	97.7	0.8	1.1	0.3	69.3	30.7	6,497	14.16		
12	71-43-2	Benzene	●●●●●●	142	no	yes	$58 \cdot 10^{-3}$	$56 \cdot 10^{-3}$	95.4	0.1	4.1	0.3	91.6	8.4	2,102	14.02		
13	95-63-6	Trimethylbenzene, 1,2,4-	●●●●●●	129	no	no	1	$300 \cdot 10^{-3}$	99.6	0.1	0.1	0.2	87.7	12.3	508	14.01		
14	75-21-8	Ethylene oxide	●●●●●●	10	no	no	2	2	99.9		0.1		99.4	0.6	38	13.95		
15	7664-41-7	Ammonia	●●●●●●	340	no	no		$18 \cdot 10^{-3}$	56.7	19.6	22.9	0.9	96.8	3.2	32,203	13.73		
16	NA - 03	Cadmium compounds	●●●●●●	44	no	yes	1	13	76.8	8.1		15.1	2.5	97.5	46	13.67		
17	79-01-6	Trichloroethylene	●●●●●●	73	no	yes	$71 \cdot 10^{-3}$	$71 \cdot 10^{-3}$	100.0				93.3	6.7	742	13.51		
18	1332-21-4	Asbestos (friable)	●●●●●●	85	yes	no		$1 \cdot 10^{+3}$				100.0	43.8	56.2	1,406	13.34		
19	91-20-3	Naphthalene	●●●●●●	59	no	no	$25 \cdot 10^{-3}$	$600 \cdot 10^{-3}$	98.3	0.2	0.1	1.4	52.3	47.7	76	13.28		
20	106-99-0	Butadiene, 1,3-	●●●●●●	16	no	no	2	2	100.0				78.7	21.3	185	13.28		
21	7550-45-0	Titanium tetrachloride	●●●●●●	3	no	no	100	100	100.0				79.9	20.1	1	13.27		
22	108-88-3	Toluene	●●●●●●	435	no	no	$2.5 \cdot 10^{-3}$	$4.5 \cdot 10^{-3}$	99.0	0.1	0.7	0.2	70.8	29.2	6,985	13.18		
23	75-07-0	Acetaldehyde	●●●●●●	41	no	no	$200 \cdot 10^{-3}$	$200 \cdot 10^{-3}$	82.8	3.3	13.8		99.1	0.9	344	13.04		
24	26471-62-5	Toluene diisocyanate (m.i.)	●●●●●●	24	no	no	$78 \cdot 10^{-3}$	26	100.0				1.9	98.1	†	12.88		
25	50-00-0	Formaldehyde	●●●●●●	133	no	no	$2.5 \cdot 10^{-3}$	$93 \cdot 10^{-3}$	83.1	13.1	3.7	0.1	84.0	16.0	1,279	12.62		
26	79-06-1	Acrylamide	●●●●●●	7	no	no	9	9.3	32.1	45.5		22.5	72.0	28.0	3	12.51		
27	75-01-4	Vinyl chloride	●●●●●●	9	yes	no	3.8	3.8	99.3	0.7			99.0	1.0	26	12.51		
28	75-09-2	Dichloromethane	●●●●●●	92	no	yes	$15 \cdot 10^{-3}$	$3.4 \cdot 10^{-3}$	100.0				93.2	6.8	2,165	12.46		
29	7429-90-5	Aluminum (fume or dust)	●●●●●●	63	no	no	$5 \cdot 10^{-3}$	$360 \cdot 10^{-3}$	6.5	11.0		82.4	7.0	93.0	737	12.38		
30	74-83-9	Bromomethane (methyl bromide)	○●●●●○	2	no	no	$360 \cdot 10^{-3}$	$360 \cdot 10^{-3}$	100.0				100.0		11	12.07		

Note: The 'CAS' column provides Chemical Abstract Service codes that uniquely identify chemicals. 'Schedule 1' and 'Toxic' are priority substances identified through the 1999 Canadian Environmental Protection Act. 'Schedule 1' substances underwent review before they were declared toxic, while 'Toxic' substances were identified in the Act but were not subject to the review process. CHHI is the EPA Chronic Human Health Indicator, expressed in Methanol-equivalent units. 'Reports' indicates if positive emissions were reported in a given report year, starting with 1993. A ○ indicates zero reports, and a ● indicates positive reports. J^+ is the number of facilities reporting positive emissions of this pollutant. The emission stream decomposition (air, water, subsoil, land) is relative to the onsite emissions. Column 'Emissions' reports the average tonnage of onsite emissions per year (unadjusted for toxicity), and column 'Exposure' shows the corresponding environmental exposure aggregate.

Table 2: Environmental Exposure Ranking — CEPA-Identified Substances

					CEPA		EPA-CHHI		Emission Stream Composition [%]						Emission	Exposure
CAS	Name	Map	<i>N</i>	Sch.1	Toxic	oral	inhale	Air	Water	Subsoil	Land	Onsite	Offsite	tons/yr.	ln <i>X</i> _{<i>c</i>}	
1	NA - 08	Lead compounds	●●●●●●	221	yes	no	100	100	41.9	5.5		52.6	27.3	72.7	1,589	20.03
4	NA - 04	Chromium Compounds	●●●●●●	314	no	yes	170·10 ⁻³	86	2.7	3.0		94.3	22.4	77.6	932	16.80
5	NA - 02	Arsenic compounds	●●●●●●	85	no	yes	3	31	5.8	0.7	90.0	3.5	96.0	4.0	2,362	16.73
8	NA - 11	Nickel compounds	●●●●●●	227	no	yes	25·10 ⁻³	3.5	68.2	8.4		23.5	56.0	44.0	701	15.61
12	71-43-2	Benzene	●●●●●●	142	no	yes	58·10 ⁻³	56·10 ⁻³	95.4	0.1	4.1	0.3	91.6	8.4	2,102	14.02
16	NA - 03	Cadmium compounds	●●●●●●	44	no	yes	1	13	76.8	8.1		15.1	2.5	97.5	46	13.67
17	79-01-6	Trichloroethylene	●●●●●●	73	no	yes	71·10 ⁻³	71·10 ⁻³	100.0				93.3	6.7	742	13.51
18	1332-21-4	Asbestos (friable)	●●●●●●	85	yes	no		1·10 ⁺³			100.0		43.8	56.2	1,406	13.34
27	75-01-4	Vinyl chloride	●●●●●●	9	yes	no	3.8	3.8	99.3	0.7			99.0	1.0	26	12.51
28	75-09-2	Dichloromethane	●●●●●●	92	no	yes	15·10 ⁻³	3.4·10 ⁻³	100.0				93.2	6.8	2,165	12.46
37	127-18-4	Tetrachloroethylene	●●●●●●	50	no	yes	50·10 ⁻³	50·10 ⁻³	99.9				66.1	33.9	158	11.30
44	NA - 10	Mercury compounds	●●●●●●	20	yes	no	6	6	92.4	6.1		1.5	3.4	96.6	3	10.83
55	56-23-5	Carbon tetrachloride	●●●●●●	11	yes	no	710·10 ⁻³	110·10 ⁻³	83.1	16.9			33.1	66.9	10	9.67
56	117-81-7	Di(2-ethylhexyl) phthalate	●●●●●●	46	no	yes	28·10 ⁻³	28·10 ⁻³	44.4			55.6	39.3	60.7	41	9.55
60	107-06-2	Dichloroethane, 1,2-	●●●●●●	10	no	yes	180·10 ⁻³	190·10 ⁻³	96.2	1.3		2.5	92.0	8.0	16	9.26

Note: The 'CAS' column provides Chemical Abstract Service codes that uniquely identify chemicals. 'Schedule 1' and 'Toxic' are priority substances identified through the 1999 Canadian Environmental Protection Act. 'Schedule 1' substances underwent review before they were declared toxic, while 'Toxic' substances were identified in the Act but were not subject to the review process. CHHI is the EPA Chronic Human Health Indicator, expressed in Methanol-equivalent units. 'Report' indicates if positive emissions were reported in a given report year, starting with 1993. A ○ indicates zero reports, and a ● indicates positive reports. *N* is the number of facilities reporting positive emissions of this pollutant. The emission stream decomposition (air, water, subsoil, land) is relative to the onsite emissions. Column 'Emissions' reports the average tonnage of onsite emissions per year (unadjusted for toxicity), and column 'Exposure' shows the corresponding environmental exposure aggregate.

Table 3: Environmental Exposure Ranking by Plant

Rank			ID	Company: Facility	Location	Emission tons	Exposure ln X
a	b	c					
1	6	8	1651	Gerdau Mrm Steel Inc.: <i>Mrm Steel</i>	Selkirk, MB	54,783	19.34
2	1	1	3824	Co-Steel Lasco:	Whitby, ON	41,708	19.31
3	2	2	1106	AltaSteel Ltd.: <i>AltaSteel Ltd</i>	Edmonton, AB	25,847	19.22
4	3	3	0444	Inco Limited: <i>Copper Cliff Smelter Complex</i>	Copper Cliff, ON	59,623	19.21
5	5	5	3623	Noranda Inc: <i>Fonderie Horne</i>	Rouyn-Noranda, QC	89,756	18.79
6	11	19	3802	Cominco Ltd.: <i>Trail Operations</i>	Trail, BC	27,281	18.63
7	4	4	3649	Ispat Sidbec Inc.: <i>Acierie</i>	Contrecoeur, QC	82,808	18.32
8	7	14	3414	Hudson Bay Mining and Smelting Company L: <i>Hbm+S Co., Ltd. - Metallurgical Complex</i>	Flin Flon, MB	62,113	18.15
9	12	13	3571	Communauté urbaine de Montréal: <i>Station d'épuration des eaux usées</i>	Montréal, QC	1,070	17.79
10	19	26	5246	Gerdau Mrm Steel Inc.: <i>Mandak Metal Processors</i>	Selkirk, MB	18,538	17.50
11	13	12	1467	Inco Limited: <i>Copper Cliff Nickel Refinery</i>	Copper Cliff, ON	9,120	17.34
12	24	25	4221	Miramar Giant Mine Limited: <i>Giant Mine</i>	Yellowknife, NT	23,012	17.30
13	14	6	3655	Ispat Sidbec Inc.: <i>Sidbec-Feruni (Ispat) Inc. Contrecoeur</i>	Contrecoeur, QC	43,042	17.29
14	16	16	4739	Dominion Castings Ltd.:	Hamilton, ON	1,112	17.27
15	15	11	2256	Tonolli Canada Limited: <i>Tonolli Canada Ltd.</i>	Mississauga, ON	861	17.22
16	10	10	4723	Vancouver Wharves Limited Partnership: <i>Vancouver Wharves Limited</i>	North Vancouver, BC	1,108	17.19
17	18	18	4526	Menasco Aerospace: <i>Bf Goodrich Aerospace Landing Gear Divis</i>	Oakville, ON	1,860	17.13
18	20	15	4435	City of Toronto: <i>Highland Creek Treatment Plant</i>	Toronto, ON	981	17.11
19	22	21	0158	Canada Metal (Est) Ltee: <i>Usine St-Léonard</i>	St-Léonard, QC	310	17.04
20	25	22	3916	Noranda inc.: <i>Affinerie Ccr</i>	Montréal-Est, QC	441	17.03
21	26	32	1197	Domtar Papers: <i>Cornwall Business Unit</i>	Cornwall, ON	3,054	17.02
22	9	9	0732	Metalex Products Ltd.:	Richmond, BC	1,079	16.89
23	23	24	3713	Dofasco Inc.: <i>Dofasco Hamilton</i>	Hamilton, ON	724	16.84
24	17	20	2161	Slater Steels Inc.: <i>Hamilton Specialty Bar Division</i>	Hamilton, ON	895	16.84
25	21	17	2984	Stelco Inc.: <i>Hilton Works</i>	Hamilton, ON	1,238	16.68
26	29	30	0930	Bowater Pulp and Paper Canada Inc.: <i>Thunder Bay Operations</i>	Thunder Bay, ON	3,144	16.45
27	27	23	2844	Ontario Power Generation Inc.: <i>Lakeview Gs</i>	Mississauga, ON	668	16.03
28	30	27	4204	Sydney Steel Corporation: <i>Sydney Steel Corp.</i>	Sydney, NS	14,511	15.99
29	8	7	2537	Safety-Kleen Ltd.: <i>Lambton Facility</i>	Corunna, ON	16,011	15.98
30	33	34	3998	Nova Scotia Power Inc.: <i>Tufts Cove Generating Station</i>	Dartmouth, NS	616	15.94

Note: The first three columns 'a,' 'b,' and 'c' indicate the rank of a facility based on its environmental exposure index, where 'a' corresponds to the narrow (2km) radius, medium (11km) radius, and wide (14km) radius. Column 'Emissions' reports the average tonnage of onsite emissions per year (expressed in toxicity-adjusted methanol-equivalent units), and column 'Exposure' shows the corresponding environmental exposure aggregate. Province codes are: AB–Alberta, BC–British Columbia, MB–Manitoba, NS–Nova Scotia, NT–Northwest Territory, ON–Ontario, QC–Québec.

Table 4: Emission Level Regression Estimates (All Pollutants)

		(A)	(B)	(C)	(D)	(E)
Employment	$\beta = \tilde{\beta}/(1 - \tilde{\gamma})$	0.247 ^c (7.824)	0.178 ^c (9.372)	0.181 ^c (9.471)	0.184 ^c (9.452)	0.289 ^c (9.635)
Environmental Exposure	$\gamma = \tilde{\gamma}/(1 - \tilde{\gamma})$	0.436 ^c (9.598)	-0.102 ^c (21.97)	-0.099 ^c (18.98)	-0.086 ^c (13.65)	0.460 ^c (15.74)
Abatement Ladder Index	δ					-0.076 ^c (33.92)
Log Likelihood	$\ln \mathcal{L}$	-24330	-24196	-24261	-24353	-23192
Observations		19,675	19,675	19,675	19,675	19,675
Fixed Effects		5,282	5,282	5,282	5,282	5,282
Facilities	J	1482	1482	1482	1482	1482
Chemicals	K	125	125	125	125	125

Note: T-statistics are shown in parenthesis. The estimates and their t-statistics shown in the table were obtained through application of the Delta method to the raw estimates (indicated by a tilde above the symbol). The transformations are $\beta = \tilde{\beta}/(1 - \tilde{\gamma})$ and $\gamma = \tilde{\gamma}/(1 - \tilde{\gamma})$. Statistical significance at the 95%, 99% and 99.9% level is indicated by the superscripts ^a, ^b, and ^c, respectively.

Table 5: Pollutant-by-Pollutant Rate-of-Change Regression Estimates, Parametric Treatment

Rank	CAS	Chemical Name	Obs.	$\tilde{\alpha}$	$\tilde{\beta}$	$\tilde{\gamma}$	δ	$\tilde{\eta}$
1	NA - 08	Lead compounds	378	0.004 (.034)	0.447 ^c (3.53)	0.025 (1.67)	0.119 (.511)	2.509 (.620)
2	7664-93-9	Sulfuric acid	365	0.190 (1.79)	0.092 (.174)	0.067 ^c (5.39)	0.139 (1.78)	-0.240 ^c (6.89)
4	NA - 04	Chromium Compounds	280	0.445 ^b (2.96)	-1.309 ^c (4.76)	0.068 ^c (3.55)	-0.044 (.860)	0.608 ^c (13.1)
5	NA - 02	Arsenic compounds	101	0.191 (1.57)	0.024 (.036)	0.029 (1.18)	0.113 (.474)	0.682 (1.96)
6	NA - 09	Manganese compounds	386	0.189 (1.05)	0.439 ^c (5.25)	0.043 ^b (2.71)	0.083 (.594)	0.398 ^c (20.8)
7	111-42-2	Diethanolamine	187	0.139 (.885)	-0.135 (.144)	0.090 ^c (3.64)	0.079 (.567)	0.407 ^c (7.23)
8	NA - 11	Nickel compounds	254	0.400 ^b (2.92)	-0.282 (1.63)	0.073 ^c (3.98)	-0.000 (.004)	0.084 ^a (2.52)
10	NA - 05	Cobalt compounds	54	0.256 (.386)	0.384 (.748)	0.051 (.715)	-0.117 (.781)	0.256 ^a (2.39)
11	7647-01-0	Hydrochloric acid	340	0.605 ^b (3.21)	0.912 ^c (3.69)	0.075 ^c (4.74)	0.001 (.037)	1.826 (.332)
12	71-43-2	Benzene	446	0.026 (.355)	-0.025 (.087)	0.041 ^c (3.53)	0.400 (1.20)	-1.963 (.747)
13	95-63-6	Trimethylbenzene, 1,2,4-	330	0.088 (1.37)	-0.036 (.159)	0.047 ^c (3.46)	1.003 (1.10)	-0.682 (1.26)
14	75-21-8	Ethylene oxide	37	1.322 (1.25)	0.373 (.273)	0.247 ^b (3.44)	0.069 (.494)	-29.24 (1.16)
15	7664-41-7	Ammonia	521	0.380 ^a (2.48)	0.071 (.454)	0.056 ^c (4.40)	0.068 (.937)	0.167 ^c (9.71)
16	NA - 03	Cadmium compounds	50	-0.025 (.048)	-0.096 (.098)	0.022 (.383)	0.044 (.065)	0.433 ^c (4.13)
17	79-01-6	Trichloroethylene	146	1.015 ^c (6.83)	0.322 (.893)	0.091 ^c (7.60)	-0.008 (.199)	-5.004 (1.10)
19	91-20-3	Naphthalene	166	0.007 (.092)	0.078 (.237)	0.127 ^c (3.75)	18.946 (.608)	-1.862 (1.24)
20	106-99-0	Butadiene, 1,3-	53	-0.041 (.229)	-0.011 (.010)	0.092 ^b (3.19)	134.12 (.001)	3.707 (.488)
22	108-88-3	Toluene	1250	0.099 ^a (2.45)	0.049 (.456)	0.030 ^c (4.14)	0.362 (1.90)	1.457 ^c (3.37)
23	75-07-0	Acetaldehyde	42	-0.193 (.516)	0.179 (.161)	-0.013 (.350)	0.229 (.081)	9.316 (.504)
24	26471-62-5	Toluene diisocyanate (m.i.)	45	0.030 (.106)	-0.379 (.559)	-0.016 (.131)	2.008 (.047)	-2.296 (.190)
25	50-00-0	Formaldehyde	293	0.398 (1.58)	0.245 (.714)	0.062 ^b (2.67)	0.166 (.908)	1.106 ^c (6.77)
27	75-01-4	Vinyl chloride	25	0.456 (.306)	5.483 ^c (4.86)	0.091 (.636)	0.030 (.102)	-11.84 (.468)
28	75-09-2	Dichloromethane	207	0.444 (1.67)	-0.281 (.684)	0.041 ^a (2.27)	-0.029 (.478)	-1.568 (.477)
29	7429-90-5	Aluminum (fume or dust)	82	1.783 ^c (5.08)	1.542 ^c (6.22)	0.171 ^c (6.46)	-0.062 (1.42)	0.912 ^c (3.98)
32	78-93-3	Methyl ethyl ketone	546	0.244 (1.95)	0.086 (.575)	0.031 ^b (2.94)	0.158 (.851)	0.135 (.076)
33	NA - 01	Antimony compounds	38	0.385 (1.09)	-1.107 (.654)	0.071 (1.56)	0.300 (.648)	29.111 (1.38)
34	NA - 06	Copper compounds	509	0.174 (1.48)	-0.118 (.334)	0.050 ^c (4.41)	0.061 (.774)	-3.012 (.738)
35	67-56-1	Methanol	874	0.434 ^c (6.28)	0.182 (1.56)	0.060 ^c (6.06)	0.092 (1.41)	2.790 (1.08)

Note: The pollutants are shown in descending order of their environmental exposure. T-statistics are shown in parenthesis. Parameter estimates have not been transformed through application of the Delta method. Statistical significance at the 95%, 99% and 99.9% level is indicated by the superscripts ^a, ^b, and ^c, respectively.

Table 6: Pollutant-by-Pollutant Rate-of-Change Regression Estimates, Non-Parametric Treatment

Rank	CAS	Chemical Name	Obs.	$\hat{\gamma} \xi \geq 0$	$\hat{\gamma} \xi \geq 0.25$	$\hat{\gamma} \xi \geq 0.5$	$\hat{\gamma} \xi \geq 0.75$	$\bar{\xi}$
1	NA - 08	Lead compounds	378	0.020 (1.75)	0.007 (.511)	0.014 (.863)	0.024 (1.32)	0.00 ^c (3.46)
2	7664-93-9	Sulfuric acid	365	0.055 ^c (5.14)	0.046 ^c (3.66)	0.060 ^c (4.01)	0.060 ^c (4.01)	0.00 ^c (13.2)
4	NA - 04	Chromium Compounds	280	0.069 ^c (3.87)	−0.001 (.015)	0.004 (.064)	0.004 (.064)	0.00 ^c (12.3)
5	NA - 02	Arsenic compounds	101	0.022 (1.51)	0.015 (.602)	0.007 (.246)	0.007 (.246)	0.00 ^c (1.82)
6	NA - 09	Manganese compounds	386	0.042 ^a (2.48)	0.012 (.989)	0.019 (1.51)	0.019 (.640)	0.00 ^c (7.35)
7	111-42-2	Diethanolamine	187	0.079 ^c (3.57)	0.043 (1.95)	0.054 ^a (2.36)	0.068 ^b (2.78)	0.00 ^c (9.90)
8	NA - 11	Nickel compounds	254	0.073 ^c (4.47)	0.020 (1.11)	0.023 (.917)	−0.003 (.053)	0.00 ^c (14.3)
10	NA - 05	Cobalt compounds	54	0.035 (.452)	−0.010 (.258)	−0.021 (.519)	0.005 (.109)	0.00 ^c (.739)
11	7647-01-0	Hydrochloric acid	340	0.075 ^c (4.77)	0.023 (1.70)	0.033 ^a (2.19)	0.024 (1.51)	0.00 ^c (16.6)
12	71-43-2	Benzene	446	0.027 ^b (2.64)	0.034 ^c (3.63)	0.029 ^b (2.69)	0.029 ^b (2.63)	0.25 ^c (17.0)
13	95-63-6	Trimethylbenzene, 1,2,4-	330	0.016 (1.28)	0.030 ^b (3.18)	0.032 ^b (3.28)	0.037 ^c (3.47)	0.75 ^c (12.5)
14	75-21-8	Ethylene oxide	37	0.237 ^b (3.43)	0.077 (1.63)	0.097 ^b (2.96)	0.097 ^b (2.96)	0.00 ^c (11.8)
15	7664-41-7	Ammonia	521	0.055 ^c (4.14)	0.013 (1.15)	0.014 (1.09)	0.004 (.154)	0.00 ^c (19.6)
16	NA - 03	Cadmium compounds	50	0.023 (.439)	0.001 (.023)	0.011 (.502)	0.002 (.085)	0.00 ^c (.263)
17	79-01-6	Trichloroethylene	146	0.090 ^c (7.54)	0.022 ^a (1.99)	0.010 (.652)	0.007 (.359)	0.00 ^c (17.6)
19	91-20-3	Naphthalene	166	0.072 ^b (2.77)	0.042 ^a (2.24)	0.050 ^b (2.82)	0.069 ^b (2.97)	0.75 ^c (8.92)
20	106-99-0	Butadiene, 1,3-	53	0.108 (1.41)	0.070 ^a (2.34)	0.070 ^a (2.34)	0.070 ^a (2.34)	0.25 ^c (4.90)
22	108-88-3	Toluene	1250	0.015 ^b (2.90)	0.014 ^b (2.90)	0.015 ^a (2.58)	0.011 (.995)	0.25 ^c (13.0)
23	75-07-0	Acetaldehyde	42	−0.008 (.170)	−0.014 (.456)	0.004 (.089)	0.004 (.099)	0.25 ^c (.429)
24	26471-62-5	Toluene diisocyanate (m.i.)	45	−0.003 (.041)	−0.014 (.259)	−0.017 (.297)	−0.020 (.343)	0.75 ^c (.448)
25	50-00-0	Formaldehyde	293	0.067 ^b (2.63)	0.033 ^b (3.07)	0.028 ^a (2.32)	0.028 ^a (2.12)	0.00 ^c (10.3)
27	75-01-4	Vinyl chloride	25	0.097 (.772)	0.029 (.559)	0.029 (.559)	−0.022 (.368)	0.00 ^c (1.65)
28	75-09-2	Dichloromethane	207	0.034 ^a (2.09)	0.003 (.194)	0.011 (.590)	0.001 (.031)	0.00 ^c (3.00)
29	7429-90-5	Aluminum (fume or dust)	82	0.144 ^c (5.34)	0.011 (.193)	0.015 (.256)	0.012 (.195)	0.00 ^c (10.6)
32	78-93-3	Methyl ethyl ketone	546	0.032 ^b (2.68)	0.010 (1.48)	0.005 (.395)	0.000 (.013)	0.00 ^c (5.49)
33	NA - 01	Antimony compounds	38	0.051 (1.21)	0.056 (1.49)	0.056 (1.49)	−0.004 (.088)	0.25 ^c (3.39)
34	NA - 06	Copper compounds	509	0.045 ^c (4.01)	0.006 (.528)	0.014 (1.04)	0.022 (1.61)	0.00 ^c (15.8)
35	67-56-1	Methanol	874	0.051 ^c (6.45)	0.023 ^b (2.60)	0.017 (1.09)	0.023 (.720)	0.00 ^c (31.7)

Note: The pollutants are shown in descending order of their environmental exposure. T-statistics are shown in parenthesis for the four γ estimates. Parameter estimates have not been transformed through application of the Delta method. The χ^2 statistics of a likelihood-ratio test is shown in parenthesis for the optimal $\bar{\xi}$, comparing the restricted case ($\xi = 1$) against the unrestricted case ($\bar{\xi} < 1$). Statistical significance at the 95%, 99% and 99.9% level is indicated by the superscripts ^a, ^b, and ^c, respectively.

Table 7: Rate-of-Change Regression Estimates (All Pollutants)

		(A)	(B)	(C)	(D)
Intercept	$\alpha = \tilde{\alpha}/(1 - \tilde{\eta})$	0.270 ^c (11.8)	0.222 ^c (9.76)	-0.111 ^c (3.41)	-0.232 ^c (9.54)
Employment Change	$\beta = \tilde{\beta}/(1 - \tilde{\eta})$	0.106 ^c (3.38)	0.111 ^c (3.51)	0.098 ^b (3.13)	0.102 ^c (3.30)
Env. Exposure	$\gamma = \tilde{\gamma}/(1 - \tilde{\eta})$	0.044 ^c (16.6)	0.051 ^c (17.5)		
Emission	$\gamma_1 = \tilde{\gamma}_1/(1 - \tilde{\eta})$			0.029 ^c (9.18)	0.017 ^c (9.71)
Toxicity	$\gamma_2 = \tilde{\gamma}_2/(1 - \tilde{\eta})$			0.029 ^c (9.18)	0.017 ^c (9.71)
Population	$\gamma_3 = \tilde{\gamma}_3/(1 - \tilde{\eta})$			0.004 (1.22)	-0.002 (.882)
Exposure Index	δ		0.127 ^c (5.26)		-0.120 ^c (21.7)
Env. Exposure Change	$\eta = \tilde{\eta}/(1 - \tilde{\eta})$	0.393 ^c (26.7)	0.392 ^c (26.6)	0.396 ^c (26.8)	0.394 ^c (27.1)
Observations		14403	14403	14403	14403
Facilities	J	1214	1214	1214	1214
Chemicals	K	109	109	109	109
Log Likelihood	$\ln \mathcal{L}$	-19950	-19919	-19818	-19716

Note: T-statistics are shown in parenthesis. Statistical significance at the 95%, 99% and 99.9% level is indicated by the superscripts ^a, ^b, and ^c, respectively. The estimates and their t-statistics shown in the table were obtained through application of the Delta method to the raw estimates (indicated by a tilde above the symbol). The transformation is shown in the table.

Table 8: Rate-of-Change Regression Estimates (CEPA Pollutants Only)

		(A)	(B)	(C)	(D)
Intercept	$\alpha = \tilde{\alpha}/(1 - \tilde{\eta})$	0.206 ^b (2.95)	0.170 ^a (2.34)	-0.080 (.742)	-0.349 ^c (6.17)
Employment Change	$\beta = \tilde{\beta}/(1 - \tilde{\eta})$	0.231 ^c (3.34)	0.238 ^c (3.41)	0.226 ^b (3.26)	0.212 ^b (3.17)
Env. Exposure	$\gamma = \tilde{\gamma}/(1 - \tilde{\eta})$	0.050 ^c (5.69)	0.053 ^c (5.62)		
Emission	$\gamma_1 = \tilde{\gamma}_1/(1 - \tilde{\eta})$			0.037 ^c (3.60)	0.011 ^b (2.69)
Toxicity	$\gamma_2 = \tilde{\gamma}_2/(1 - \tilde{\eta})$			0.037 ^c (3.60)	0.011 ^b (2.69)
Population	$\gamma_3 = \tilde{\gamma}_3/(1 - \tilde{\eta})$			0.009 (.668)	-0.005 (1.71)
Exposure Index	δ		0.058 (1.08)		-0.186 ^c (9.43)
Env. Exposure Change	$\eta = \tilde{\eta}/(1 - \tilde{\eta})$	0.714 ^c (15.1)	0.717 ^c (15.2)	0.737 ^c (15.0)	0.731 ^c (15.5)
Observations		2077	2077	2077	2077
Facilities	J	394	394	394	394
Chemicals	K	15	15	15	15
Log Likelihood	$\ln \mathcal{L}$	-2820	-2819	-2805	-2771

Note: T-statistics are shown in parenthesis. Statistical significance at the 95%, 99% and 99.9% level is indicated by the superscripts ^a, ^b, and ^c, respectively. The estimates and their t-statistics shown in the table were obtained through application of the Delta method to the raw estimates (indicated by a tilde above the symbol). The transformation is shown in the table.